

What Are Intangibles Worth?

Production versus Market Power

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Abstract

Intangibles create firm value by raising productivity and strengthening market power. The former boosts growth, whereas the latter may weaken competition. I quantify both channels in a dynamic model of intangible investment and decompose intangible value for each U.S. public firm. By 2024, market power accounts for more aggregate market capitalization than production—28% versus 21%. Intangible value grows as within-firm rents rise and firms with larger productivity gains capture market share. Over the firm life cycle, intangible value shifts from production toward market power. Intangible-driven market power helps explain the declining value premium and firms' incentives for intangible investments.

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1 Introduction

The rise of intangible capital has transformed both how firms produce and how they compete. Investments in R&D, software, and organizational know-how raise productivity, but they also strengthen market power by differentiating products and protecting firms from competition (Romer, 1990; De Ridder, 2024). While a large literature focuses on the total contribution of intangibles to firm valuation and investment (e.g., Peters and Taylor, 2017; Belo et al., 2022), much less is known about the relative importance of production versus market power channels. Distinguishing these two channels is economically important. Productivity gains promote growth, whereas market power may weaken competition and reduce investment incentives. This paper separates and quantifies these two sources of intangible value.

I construct a dynamic model in which intangible investment both enhances productivity and generates market power. Combining the estimated model with data on U.S. firms, I decompose the market value for each firm-year into four components: the value of physical capital, intangible capital in production, intangible capital in market power, and market power unrelated to intangibles. I then examine how the two economic roles of intangibles have evolved and how each separately affects firm value and investment decisions.

I begin by constructing a model that links intangible capital endogenously to market power. Three features distinguish it from the standard neoclassical framework. First, intangible capital raises markups, increasing the share of rents in profits and firm value. Second, firms face idiosyncratic markup shocks from changes in the competitive environment and other non-intangible forces, such as shifts in consumer preferences or rivals' strategies. Together, these two sources of market power introduce imperfect competition and non-constant returns to scale in profits. Third, firms differ in how efficiently they use physical and intangible capital, which generates additional variation in firm value.

I estimate the model using Simulated Method of Moments (SMM) and target key empirical moments from U.S. public firms over 1976–2024. I then combine the model's value function with state variables extracted from the data to recover model-implied value for each

firm-year. The model fits the data well and explains over 80% of the cross-sectional variation in valuation. Next, I apply the Shapley decomposition method (Shapley, 1951) to quantify the marginal contribution of each state variable and disentangle the dual roles of intangibles.

I find that, on average, the value of intangibles in market power exceeds their value in production, accounting for 28.2% and 16.5% of firm value, respectively. I validate the interpretation of the intangible market power component using patent events, which provide observable changes in firms' innovative assets. Following a firm's first highly cited patent, the value share of intangible-driven market power rises, while non-intangible market power remains largely unchanged. As a complementary test, exposure to the China tariff shock affects non-intangible market power but not the intangible-driven component. Together, these results support the model's ability to isolate the value of intangible market power and suggest that the non-intangible component plausibly captures the external competitive environment.

The value and composition of intangibles vary significantly over time, across the firm life cycle, and in the cross section. Over time, the value share of intangibles in production rises from 12.5% in the first half to 20.6% in the second half of the sample, largely because revenues shift toward more intangible-intensive firms. In contrast, intangible-driven market power remains stable around 28%, suggesting that the market power channel has been persistently important. Non-intangible market power also rises, almost entirely because firms with greater market power capture larger revenue shares. This pattern is consistent with weaker competition in the economy and the growing market shares of firms facing less competition (Grullon, Larkin and Michaely, 2019; De Loecker, Eeckhout and Unger, 2020).

As firms mature, the production value of intangibles declines while the market power contribution rises, suggesting that young firms initially benefit from proprietary technologies in production and increasingly monetize them through market power. Firms tend to enter under favorable competitive conditions, so non-intangible market power is initially high. As they age, they face more competition and increasingly rely on intangible-driven market power. Across firms, intangibles explain 45% of the variation in value. Moreover, as firms

increasingly differ in the strength of their market power, the two rent components together account for around 30% of cross-sectional variation by the end of the sample.

Incorporating intangible-driven market power is important for understanding both asset prices and firm investment. I first show that it helps explain the decline in the value premium (Fama and French, 2021). Book-to-market has become a noisier signal of expected returns because book equity primarily captures productive capital, whereas market equity also includes market power rents. Consistent with this mechanism, removing both intangible and non-intangible market power from the market equity yields a significant premium of 0.33% per month, whereas removing only non-intangible market power produces a small and insignificant return spread. A value signal combining book-to-market with model-implied market power shares generates a premium of 0.49% ($t = 3.13$) with no significant downward trend. These results suggest that growing cross-sectional differences in market power have weakened conventional value signals.

The value decomposition also allows me to construct a separate q for each economic component and identify the forces underlying investment. Neoclassical investment theory predicts that q summarizes a firm's investment opportunities (Hayashi, 1982), but conventional measures aggregate economically distinct sources of value. I therefore construct four component q 's from the value decomposition. Both intangible components strongly predict intangible investment, while market power provides a stronger incentive for intangible investment, with the market power q carrying a larger coefficient. By contrast, non-intangible market power q predicts neither physical nor intangible investment strongly, consistent with weaker investment incentives in less competitive environments.

This paper contributes to several literatures that, despite deep theoretical connections, have developed largely in parallel. A central insight from the endogenous growth literature is that intangible capital and market power are inherently linked. Monopoly rents are necessary to finance intangible investments, and intangibles can strengthen market power by differentiating products and deterring entry (Romer, 1990; Grossman and Helpman, 1991;

Aghion and Howitt, 1992; De Ridder, 2024). Yet the finance literature has not incorporated the endogenous link between intangible capital and market power. This paper bridges the gap with a structural model in which firms invest in intangibles to accumulate markups.

This paper contributes to the literature on the rise of intangible capital and its impact on firm valuations and investments (e.g., Peters and Taylor, 2017; Kogan et al., 2017; Falato et al., 2022). A closely related paper in this literature is Crouzet and Eberly (2023). They decompose the wedge between average and marginal q into intangibles, exogenous rents, and a nonlinear interaction term, highlighting key aggregate and sectoral patterns. My paper complements their analysis along two dimensions. First, I model the endogenous link between intangible capital and markups. Second, I provide firm-level value decompositions and focus on how the composition varies in the *cross section*.

This paper also contributes to investment-based asset pricing that studies asset prices through the lens of firm investment decisions (e.g., Merz and Yashiv, 2007; Liu, Whited and Zhang, 2009). Building on Belo, Xue and Zhang (2013), Belo et al. (2022) decompose firm value into components of physical capital, labor, and intangible inputs using a neoclassical investment model. This paper departs from the neoclassical framework by modeling non-constant returns to scale and explicitly allowing for market power. The resulting decomposition quantifies the contribution of market power separately from that of production inputs. Additionally, my framework applies SMM estimation and recovers firm-specific value decompositions, extending beyond the portfolio-level estimation in Belo et al. (2022).

This paper is also connected to the recent literature on declining competition and rising market power. Several papers document a broad increase in market power over recent decades. For example, De Loecker, Eeckhout and Unger (2020) find that aggregate markups for U.S. public firms rose from 21% to 61% above marginal cost over the past four decades. Grullon, Larkin and Michaely (2019) show that industries have become more concentrated. Market power also affects firm risk and expected returns (Bustamante and Donangelo, 2017; Corhay, Kung and Schmid, 2020; Dou, Ji and Wu, 2021) and helps explain the growing

divergence between small and superstar firms (Corhay, Kung and Schmid, 2025). This paper instead examines the sources of market power and shows that the rise in aggregate market power is driven mainly by non-intangible forces, consistent with declining competition.

The rest of the paper is organized as follows. Section 2 presents motivating facts and theories on the relation between intangible capital and market power. Section 3 describes a dynamic firm model with intangible capital and endogenous market power. Section 4 outlines the estimation and the decomposition method. Section 5 presents key quantitative results on value decomposition. Section 6 presents two applications: the decline in the value premium and the investment- q relation. Finally, Section 7 concludes.

2 Motivation

The relation between intangible capital and market power is pronounced in the data and is also well-established in economic theory. Section 2.1 presents motivating evidence on the relation between intangibles and markups, including the cross-sectional correlation and the markup response after filing and issuance of influential patents. Section 2.2 discusses the economic mechanism through which intangibles generate market power.

2.1 Empirical Facts

2.1.1 Data

I use accounting data on publicly traded firms from the Compustat annual file and market equity from the Center for Research in Security Prices (CRSP) monthly file. I keep only firms with a share code (SHRCD) of 10 or 11 and exchange code (EXCHCD) of 1, 2, or 3. I also exclude firms with non-positive sales, book assets, or firm value and firms with less than \$5 million in physical capital (Compustat item PPEGT, deflated in 1990 dollars). I exclude firms from regulated utilities (SIC codes 4900–4999), financial firms (6000–6999), and firms categorized as public service, international affairs, or nonoperating establishments (9000+).

I use the measure of intangible capital from Peters and Taylor (2017), which capitalizes R&D and 30% of SG&A expenses along with balance-sheet goodwill. I exclude firms with missing or zero intangible stock. The final sample contains 122,701 firm-year observations from 1976 to 2024, which correspond to 11,272 distinct firms. The sample starts in 1976 because it is the first year when the intangible capital measure is available.

In this section, I measure market power using firm-level markups based on the production function approach, which recovers the markup as the ratio of a variable input’s output elasticity to its revenue share.¹ In the main text, I focus on the estimates under the Cobb-Douglas production technology with variable inputs, physical capital, and intangible capital as production inputs. This measure is consistent with the model production function in the next section. To further ensure the validity of the markup estimates, I also include five additional measures: two model-free markup proxies and three production-function-based estimates under different assumptions on the production technology. Detailed definitions and results based on these measures are provided in Internet Appendices A.1 and B.1.

2.1.2 Correlation between Intangibles and Market Power

Figure 1(a) plots the annual cross-sectional correlation between intangible capital share and markup, along with 95% confidence intervals. The correlation is consistently positive and statistically significant throughout the sample period, ranging from approximately 0.2 to 0.4. Moreover, the correlation stays positive across time and exhibits a modest upward trend, suggesting that the link between intangible intensity and market power has been a persistent phenomenon in the economy. Figure 1(b) further illustrates this relation through binned scatter plots. Both variables are calculated as residuals after purging out industry and year fixed effects to isolate within-industry variation. The relationship is positive and

¹While this widely used method provides markup estimates for nearly all public firms, the literature has raised several concerns, such as identification without quantity data and imperfect input proxies (Bond et al., 2021; Demirer, 2022; Raval, 2023; Traina, 2018). However, De Ridder, Grassi and Morzenti (2026) argue that, although possibly biased in levels, production-based estimates accurately capture cross-sectional dispersion in markups.

approximately linear, with markups rising from roughly one at the lowest intangible share bins to close to two at the highest. Results are largely similar when using alternative markup measures and in a regression framework with firm-level controls (Internet Appendix [B.1](#)).

2.1.3 Patent Event Studies

Successful patents are an observable outcome of prior intangible investment. Patent events therefore provide a direct setting to measure the market power gains from these successful innovations. Because the importance of most innovations is difficult to observe directly, I focus on patents ultimately revealed to be influential. Specifically, I classify a patent as important if either its forward citation count or its economic value, as measured by Kogan et al. ([2017](#)), lies above the 90th percentile within its Fama-French 48 industry-year. Citations and economic value capture an innovation’s technological and economic significance, so these patents are more likely to represent successful innovations and contribute to firms’ intangible stock. I then trace the evolution of markups around the filing and grant dates of these important patents.

Figure [2](#) reports event study results around the filing (left column) and grant (right column) of each firm’s first important patent, under both definitions.² I focus on the first important patent because patents create durable market power, and earlier patents would contaminate estimates around subsequent ones. A stacked design in Internet Appendix [B.1](#) using all highly cited or high-value patents delivers similar results but exhibits pre-trends.

Markups rise by roughly 3% for highly cited patents and 8% for high-value patents around the filing year of influential patents. The larger response for high-value patents reflects the fact that the economic value measure embeds investors’ expectations of future profits and market power. Patent filing is the time when a firm has already developed an innovation ultimately revealed to be valuable. As firms commercialize these innovations, product differentiation and other competitive advantages translate into greater pricing power

²Results under alternative markup measures are reported in Internet Appendix [B.1](#).

and higher markups. Consistent with this interpretation, the response around the grant date is positive but weaker and preceded by pre-trends, indicating that markups capitalize part of the economic value of successful innovations before formal patent approval.

2.2 Why Intangibles Generate Market Power

The relation between innovation and market power is central to the endogenous growth literature. Successful innovation generates temporary monopoly rents that allow firms to recover the fixed costs of creating knowledge (Romer, 1990; Grossman and Helpman, 1991; Aghion and Howitt, 1992; Klette and Kortum, 2004). In the industrial organization literature, patents protect these rents, and firms may also strategically invest in innovation to obtain or preserve market leadership (Nordhaus, 1969; Loury, 1979; Dasgupta and Stiglitz, 1980). In these models, intangible inputs are modeled as disembodied ideas rather than as firms' own assets. However, the same logic carries over to dynamic firm models in which innovations are capitalized into an intangible capital stock.

Intangibles other than innovations and patents can also create market power. Advertising, branding, and customer relationships can reduce consumers' price sensitivity by increasing switching costs or strengthening brand loyalty (Gourio and Rudanko, 2014). Investments in product development and perceived quality can further differentiate a firm's products from its competitors', allowing the firm to charge higher prices (Dixit and Stiglitz, 1977; Doraszelski and Markovich, 2007). Organizational intangibles raise fixed costs or generate scale advantages, discouraging entry and weakening competitive pressure (De Ridder, 2024).

These theoretical mechanisms suggest that accumulated intangible capital can generate firm value through market power. The empirical evidence presented in Section 2.1 is consistent with these mechanisms: firms with greater intangible capital have higher markups, and successful innovations are followed by increases in markups. These findings motivate the model developed in the next section, in which intangible capital simultaneously affects production and market power.

3 Model

Time is discrete, and the horizon is infinite. Firms use physical capital, intangible capital, and costlessly adjustable inputs to produce heterogeneous products. Firms face imperfect competition in the product market and can invest in intangibles to raise their markup. In each period, firms jointly choose the price and flexible inputs to maximize operating profits, subject to the market demand. Taking the operating profits as given, firms then choose physical and intangible investments to maximize the market value of equity.

3.1 Technology

Firm i produces under a Cobb-Douglas technology: $Y_{it} = \mathbf{K}_{it}^\eta L_{it}^{1-\eta}$, in which L_{it} is a bundle of flexible inputs, e.g., labor and materials. $\mathbf{K}_{it}$ is a CES capital aggregator that follows

$$\mathbf{K}_{it} = \mathbf{K}(z_{it}^K, z_{it}^N, K_{it}, N_{it}) = [\alpha(\exp(z_{it}^K)K_{it})^\rho + (1 - \alpha)(\exp(z_{it}^N)N_{it})^\rho]^\frac{1}{\rho}, \quad (1)$$

in which K_{it} is physical capital and N_{it} is intangible capital. The parameter ρ governs the curvature of the CES aggregator. The corresponding elasticity of substitution between physical and intangible capital is $\sigma = 1/(1 - \rho)$. Two capital-specific productivity shocks z_{it}^K and z_{it}^N capture the stochastic productivity of physical and intangible capital. They are firm-specific and thus contribute to cross-sectional heterogeneity in the model. I assume production exhibits constant returns to scale in capital and the flexible input. I model intangible capital as a single state variable N , which summarizes the firm's stock of R&D, brand capital, organizational capital, and related intangible assets. This parsimonious representation captures the common role of these assets in affecting both production and market power. Section 3.5 discusses these modeling choices in more detail.

Firms have pricing power in the product market, and markups reflect the degree of their market power. Consistent with the industrial organization literature (e.g., Tirole, 1988), the markup reflects a firm's ability to extract rents, reduce pass-through, and protect profits

from competition. Firm i charges a markup of μ_{it} , defined as price over marginal cost. μ_{it} follows the functional form below:

$$\mu_{it} = \mu(N_{it}, \zeta_{it}^\mu) = \exp(\zeta_{it}^\mu)(1 + \bar{\mu} \cdot N_{it}^\phi). \quad (2)$$

The exogenous shock ζ_{it}^μ captures the competitive environment the firm faces, and parameters $\bar{\mu}$ and ϕ govern the magnitude of intangible capital's effect on market power.

The markup specification embeds two distinct sources of market power. First, firms endogenously accumulate market power through investment in intangible capital. For example, firms with better IT systems and management practices could produce at lower costs but still charge similar prices to those charged by their less efficient rivals. Firms with high brand value can reduce customer price sensitivity and sustain higher markups over time. Second, market power may shift due to non-intangible forces, captured by the exogenous shock ζ_{it}^μ . These shocks can be either positive, reflecting favorable market conditions such as a high entry barrier or relaxed antitrust policy, or negative, reflecting increased competition or rivals' innovation. I do not allow physical capital to directly affect market power. Section 3.5 discusses this assumption, and Section 4.6 validates the economic meaning of this shock.

In each period, the firm chooses its optimal level of flexible input and price to maximize profits, taking demand and the price of the input, P_{Lit} , as given. Let $\Pi_{it} = \Pi(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it})$ denote the operating profits for firm i at time t (Appendix A provides details for derivations):³

$$\Pi_{it} = \max_{L_{it}, P_{it}} P_{it}Y_{it} - P_{Lit}L_{it} = \begin{cases} \underbrace{\left[1 - \frac{1-\eta}{\mu_{it}}\right] \left(\frac{1-\eta}{\mu_{it}}\right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} \bar{Y} \cdot \mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}}_{\bar{\Pi}(\mu_{it})} & \text{if } \mu_{it} > 1 \\ \eta(1-\eta)^{\frac{1-\eta}{\eta}} \bar{Y} \cdot \mathbf{K}_{it} & \text{if } \mu_{it} \leq 1, \end{cases} \quad (3)$$

where $\bar{Y}$ is the size of the market and is a constant. When the markup is less than or equal

³This profit function applies as long as the firm's implied revenue does not exceed the total market size $\bar{Y}$. If a firm is productive enough to hit this ceiling, it captures the entire market and its profit is governed by the capped-revenue regime detailed in Appendix A.

to 1, the firm is assumed to operate under perfect competition and take prices as given.

The profit function shows two effects of market power. On the one hand, firms earn more profits with higher price-setting power, as $\bar{\Pi}(\mu_{it})$ is increasing in μ_{it} . On the other hand, market power decreases the incentive to invest through the second term $\mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}$ as higher μ_{it} decreases the marginal benefit of capital. Moreover, for a given μ_{it} , the profit function is homogeneous of degree $\eta/(\mu_{it} - 1 + \eta)$. When the market structure approaches perfect competition ($\mu_{it} \rightarrow 1$), the profit function has constant returns to scale. As firms accumulate market power, rents constitute a larger share of firms' profits and marginal returns to capital could fall given that returns to scale $\eta/(\mu_{it} - 1 + \eta)$ are decreasing in μ_{it} . This feature is in line with the recent trend of declining capital share in the economy (Barkai, 2020).

3.2 Investment

Capital equals investment plus beginning-of-period capital net of depreciation:

$$K_{it+1} = (1 - \delta^K)K_{it} + I_{it}^K, \quad (4)$$

$$N_{it+1} = (1 - \delta^N)N_{it} + I_{it}^N, \quad I_{it}^N \geq 0. \quad (5)$$

The depreciation rate δ^N for intangible capital can be interpreted as the rate of creative destruction. New technologies render old ones obsolete, and brand loyalty wears off over time. I do not allow disinvestment in intangible capital. Intangible investments are difficult to resell in secondary markets and are therefore largely sunk and irreversible once incurred.

Investment in both types of capital incurs quadratic costs:

$$\Phi^K(I_{it}^K, K_{it}) = I_{it}^K + \frac{b^K}{2} \left(\frac{I_{it}^K}{K_{it}} \right)^2 K_{it}, \quad (6)$$

$$\Phi^N(I_{it}^N, N_{it}) = I_{it}^N + \frac{b^N}{2} \left(\frac{I_{it}^N}{N_{it}} \right)^2 N_{it}. \quad (7)$$

The quadratic adjustment cost for physical investment is standard in the q -theory literature.

The same intuition can be generalized to intangible capital. The convex adjustment cost captures diminishing returns to intangible investment as well as constant returns to scale in I^N and N jointly, which is in line with the stylized fact that the innovation rate is independent of firm size (Klette and Kortum, 2004).

3.3 Stochastic Processes and the Stochastic Discounting Factor

I jointly model aggregate and idiosyncratic productivity shocks as a single state variable, which implicitly assumes that they have the same persistence. This choice mitigates the curse of dimensionality and ensures that the structural estimation remains computationally feasible. Two capital-specific productivity shocks follow:

$$z_{it+1}^K = x_{t+1}^K + \zeta_{it+1}^K = \rho^K(x_t^K + \zeta_{it}^K) + \varepsilon_{x,t+1}^K + \varepsilon_{\zeta,it+1}^K, \quad (8)$$

$$z_{it+1}^N = x_{t+1}^N + \zeta_{it+1}^N = \rho^N(x_t^N + \zeta_{it}^N) + \varepsilon_{x,t+1}^N + \varepsilon_{\zeta,it+1}^N, \quad (9)$$

where, for $j \in \{K, N\}$, x_{t+1}^j is aggregate productivity, ζ_{it+1}^j is firm-specific productivity, and $\varepsilon_{x,t+1}^j$ and $\varepsilon_{\zeta,it+1}^j$ are i.i.d. aggregate and firm-specific shocks, respectively.

Following Zhang (2005), I specify the stochastic discount factor exogenously as:

$$M_{t+1} = \exp \left(-r_f - \gamma^K \varepsilon_{x,t+1}^K - \gamma^N \varepsilon_{x,t+1}^N - \frac{(\gamma^K)^2 (\sigma_x^K)^2}{2} - \frac{(\gamma^N)^2 (\sigma_x^N)^2}{2} \right), \quad (10)$$

where r_f , γ^K , γ^N correspond to the interest rate and the prices of risk for physical and intangible productivity shocks. σ_x^K and σ_x^N are the standard deviations of the aggregate productivity shocks for physical and intangible capital, respectively.

The exogenous shock to markup is given by

$$\zeta_{it+1}^\mu = \rho^\mu \zeta_{it}^\mu + \varepsilon_{\mu,it+1}, \quad (11)$$

in which $\varepsilon_{\mu,it+1}$ is independently and identically distributed and uncorrelated across firms.

3.4 Firms' Maximization Problem

The payout of firm i equals

$$D_{it} \equiv \Pi(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) - \Phi^K(I_{it}^K, K_{it}) - \Phi^N(I_{it}^N, N_{it}) - f, \quad (12)$$

where f is the fixed operating cost. Firm i maximizes its cum-dividend market value of equity:

$$V_{it}^{\text{cum}} = \max_{\{I_{it+s}^K, I_{it+s}^N\}_{s=0}^{\infty}} \mathbb{E}_t \left[\sum_{s=0}^{\infty} M_{t,t+s} D_{i,t+s} \right]. \quad (13)$$

Firms are characterized by the vector of state variables $(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it})$. Firms' maximization problem can then be written recursively as

$$V^{\text{cum}}(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) = \max_{K_{it+1}, N_{it+1}} D_{it} + \mathbb{E}_t [M_{t+1} V^{\text{cum}}(z_{it+1}^K, z_{it+1}^N, \zeta_{it+1}^\mu, K_{it+1}, N_{it+1})], \quad (14)$$

subject to capital accumulation equations (4) and (5). Lastly, I rewrite the value function as the ex-dividend value to match the data valuation:

$$V(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) = V^{\text{cum}}(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) - D_{it}. \quad (15)$$

3.5 Discussion of Modeling Assumptions

Constant returns-to-scale. The assumption of constant returns to scale in production is motivated by both identification and consistency with the prior literature. First, recent studies argue that returns to scale and markups generally cannot be separately identified (Flynn, Gandhi and Traina, 2019). Second, CRS is a standard assumption in a large body of empirical and theoretical work (e.g., Atkeson and Burstein, 2008; Edmond, Midrigan and Xu, 2023). Flynn, Gandhi and Traina (2019) further document that estimated returns

to scale in U.S. firm-level data are centered close to one, supporting CRS as a reasonable empirical approximation.

Intangible capital as a single state variable. While intangible capital encompasses a variety of components, I model it as a single firm-level input because this paper seeks to decompose its effects by economic channel rather than by type of intangible asset. This approach also reflects important measurement constraints. Current accounting standards do not fully recognize intangible capital on firms’ balance sheets, and measures of specific components typically do not cover the full universe of public firms. Moreover, different types of intangibles often operate through overlapping channels. For example, R&D and organizational know-how can simultaneously enhance production and strengthen market power through lower marginal costs, product differentiation, and customer loyalty. Aggregating intangible components therefore avoids multiple state variables while preserving the distinct economic mechanisms through which intangible capital affects firms. Although these effects may vary across components, the model captures their average effects and provides a parsimonious, tractable framework for estimation and decomposition.

Markup and physical capital. I do not allow physical capital to directly affect markups. This modeling choice is motivated by the empirical relationship between intangibles and markups, as well as by the economic properties of intangibles, discussed in Section 2. Other factors, including physical capital, may also contribute to market power—for example, through equipment scarcity or entry barriers. These non-intangible sources of market power are captured by the exogenous shock ζ_{it}^μ , allowing for a flexible representation of market power variation not directly tied to intangible assets.

4 Estimation and Decomposition

In this section, I first outline the strategy for identifying model parameters in Section 4.1 and show the estimation results in Section 4.2. Then I discuss the decomposition framework and the measurement of state variables in Section 4.3. Finally, Section 4.5 evaluates the model fit and Section 4.6 validates the decomposition.

4.1 Identification

There are 22 parameters in the model. I calibrate four parameters as reported in Panel A of Table 1 and estimate the remaining 18 parameters. Specifically, I set the constant annual risk-free rate r_f to be 4%, matching the data value during the sample period 1976–2024. The size of the product market, $\bar{Y}$, is set to be 20, which normalizes the long-term average level of total capital to be around four. The annual depreciation rate of physical capital, δ^K , is set to be 0.12, following Gomes (2001) and Cooper and Haltiwanger (2006). The depreciation rate for intangible capital, δ^N , is calibrated to 0.2, close to both the average rate in Peters and Taylor (2017) and the rate calibrated in Falato et al. (2022).

I estimate the remaining 18 parameters using the Simulated Method of Moments (SMM). Identification relies on selecting moments that are informative about underlying structural parameters. To discipline the estimation, I group the parameters into four categories according to the margins they primarily govern. The first group concerns the production technology. I use the mean of the profits-to-total-capital ratio to identify the share of capital in production, η . I include the average ratio of intangible capital to total capital to pin down the share of physical capital in production, α . To identify the CES substitution parameter, ρ , I rely on the standard deviation of the intangible-to-total-capital ratio. When the two capital types are complementary, intangible intensity should remain close to its optimal ratio and exhibit smaller variation in the data. Lastly, I include the average total value-to-total-capital ratio to identify the fixed operating cost f .

The second group includes parameters that govern investment in physical and intangible capital. I use the standard deviations of the physical and intangible investment rates to pin down the adjustment cost parameters, $\{b^K, b^N\}$, respectively. For the parameters that govern two productivity processes, I first match the AR(1) coefficient and the standard deviation of profits scaled by physical capital to identify the persistence and volatility of the idiosyncratic physical productivity shock, ρ^K and σ_ζ^K . For the intangible-productivity shock, I use the AR(1) coefficient and the standard deviation of profits scaled by intangible capital to identify the persistence and volatility, $\{\rho^N, \sigma_\zeta^N\}$, respectively. These two profitability moments load differently on the two shocks and help disentangle their respective contributions. Since I jointly model the persistence of aggregate and idiosyncratic productivity shocks, I additionally target the AR(1) coefficient of both firm-level and aggregate profits scaled by total capital to discipline the persistence of shocks.

The third group of parameters captures the interaction between intangible capital and markup. I use moments of the ratio between sales and cost of goods sold (Compustat item SALE/COGS) to discipline markup-related parameters. I identify the average effect of intangibles, $\bar{\mu}$, and the curvature of the markup function, ϕ , using the average SALE/COGS and the cross-sectional correlation between intangible capital and SALE/COGS. I then rely on the AR(1) coefficient of SALE/COGS to identify the persistence of exogenous markup shocks, ρ^μ , and the standard deviation of SALE/COGS to pin down their volatility, σ^μ .

The last group governs the stochastic discounting factor and the aggregate productivity. I use the average market return to identify the risk price of physical productivity shocks, γ^K , and the average return spread between high- and low-profitability firms to identify the risk price of intangible productivity shocks, γ^N . I then use the volatility of aggregate physical and intangible investment to identify the standard deviation of the two aggregate productivity shocks, $\{\sigma_x^K, \sigma_x^N\}$. I also include the volatility of aggregate profitability to help identify the volatility of these aggregate shocks.

Because the return moments are estimated much less precisely than the quantity mo-

ments, I estimate the parameters iteratively in two stages. In the first stage, I use only the quantity moments. In the second stage, holding the first-stage estimates fixed, I use two return moments to identify the prices of risk associated with physical- and intangible-productivity shocks. I alternate between the two stages until the estimates converge.

4.2 Estimation Results

Panel B of Table 1 reports the estimated parameters. Overall, the estimated parameters are economically plausible and broadly consistent with values used in the prior literature. Regarding production technology, physical capital accounts for around 73% of total capital, in line with recent studies such as Crouzet and Eberly (2023). The total capital share is around 0.27, close to that in Bloom (2009). The estimated CES curvature parameter is $\rho = 0.45$, suggesting that physical and intangible capital are more substitutable than under the Cobb-Douglas benchmark, for which $\rho = 0$. This degree of substitutability is consistent with the large dispersion in intangible intensity observed in the data and allows firms to operate with little or no intangible capital.

In the second group, the adjustment cost for physical capital is estimated to be larger than that for intangible capital, which is consistent with technological progress over recent decades as intangible investment has become less costly. In addition, physical productivity shocks are larger and less persistent than intangible productivity shocks. This difference suggests that heterogeneity in intangible productivity reflects persistent differences, whereas physical productivity is subject to more transitory shocks. The third group characterizes the behavior of markups and the interaction with intangible capital. The estimate of $\phi < 1$ implies diminishing marginal effects of intangible capital on markups. The exogenous markup shock is less persistent and less volatile than the productivity shocks, implying that market power is shaped by more transitory forces and evolves more gradually over time.

Table 2 presents the 21 moment conditions used in the estimation. Panel A reports moments based on firm-level characteristics and Panel B reports the aggregate moments.

Simulated moments tend to be reasonably close to their empirical counterparts, with some gaps, as expected given the large number of moments I consider.

4.3 Shapley Decomposition

Interactions among the state variables make their individual contributions to firm value inherently order-dependent. I therefore employ the Shapley decomposition (Shapley, 1951) to decompose firm value. The method averages each state variable's marginal contribution across all possible orderings, thereby fairly allocating firm value while accounting for interactions among the states.⁴

To fix ideas, consider a simplified example with only two state variables, physical capital K and intangible capital N . The Shapley decomposition of firm value is

$$V(K_{it}, N_{it}) = V(K_{\min}, N_{\min}) + V_{it}^K + V_{it}^N, \quad (16)$$

where $K_{\min}, N_{\min}$ correspond to the minimum grid values for the two types of capital. There are two orderings in which the two variables can be added: K enters first, followed by N ; or N enters first, followed by K . The Shapley values average the contribution of variables over these two orderings:

$$\begin{aligned} V_{it}^K &= \frac{1}{2} \left[\underbrace{V(K_{it}, N_{\min}) - V(K_{\min}, N_{\min})}_{K \text{ enters first}} + \underbrace{V(K_{it}, N_{it}) - V(K_{\min}, N_{it})}_{K \text{ enters second}} \right], \\ V_{it}^N &= \frac{1}{2} \left[\underbrace{V(K_{\min}, N_{it}) - V(K_{\min}, N_{\min})}_{N \text{ enters first}} + \underbrace{V(K_{it}, N_{it}) - V(K_{it}, N_{\min})}_{N \text{ enters second}} \right]. \end{aligned} \quad (17)$$

By construction, $V_{it}^K + V_{it}^N = V(K_{it}, N_{it}) - V(K_{\min}, N_{\min})$, so the decomposition is exact. Crucially, the Shapley approach accounts for complementarities between K and N as any interaction effect is split between the two variables, rather than being attributed entirely to

⁴Shapley decompositions have been widely used in economics and finance, including studies of income inequality (Shorrocks, 1982), worker- and firm-level wage dispersion (Abowd, Kramarz and Margolis, 1999), and heterogeneity in investor expectations (Couts, Gonçalves and Loudis, 2023).

whichever variable is added last.

Formally, I apply this same logic to the full value function, and decompose the contribution of intangible capital into production and markup channels:

$$\begin{aligned}
V_{it} &= V(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) \\
&= \underbrace{\bar{V} + V_{it}^{z^K} + V_{it}^K}_{\equiv V_{it}^{\text{phy}}} + \underbrace{V_{it}^{z^N} + V_{it}^{N,\text{prod}}}_{\equiv V_{it}^{\text{prod-int}}} + \underbrace{V_{it}^{N,\text{mkp}}}_{\equiv V_{it}^{\text{mkp-int}}} + \underbrace{V_{it}^{\zeta^\mu}}_{\equiv V_{it}^{\text{mkp-non}}} , \tag{18}
\end{aligned}$$

where $\bar{V} \equiv V(z^K = 0, z^N = 0, \zeta^\mu = 0, K = K_{\min}, N = N_{\min})$ is the value function evaluated at constant benchmark values for the five state variables. $\bar{V}$ represents the baseline value of a firm with close-to-zero levels of capital and subject to no exogenous shock. Each Shapley value term V_{it}^{comp} measures the marginal contribution of each variable to firm value, averaged over all possible orderings.

$V_{it}^{N,\text{prod}}$ and $V_{it}^{N,\text{mkp}}$ capture the contribution of intangible capital through production and markup, respectively. They cannot be separated directly in the current model because intangibles enter as a single state variable. I therefore solve an auxiliary model with separate intangible variables for production and markup. I then apply a six-variable Shapley decomposition that treats N^{prod} and N^{mkp} as distinct. I use the original value function when the two variables are equal and the auxiliary value function when only one differs. The sum of $V_{it}^{N,\text{prod}}$ and $V_{it}^{N,\text{mkp}}$ is the total contribution of the intangible capital. The details are provided in Internet Appendix [A.3](#).

I group the resulting six Shapley terms in Equation (18) into four economic components. The value of capital in production, V_{it}^{phy} and $V_{it}^{\text{prod-int}}$, contains both the capital-specific productivity and the capital stock. The value of physical capital, V_{it}^{phy} , contains the constant term $\bar{V}$ as it represents the benchmark value of a firm without intangible capital and market power. Two market power components, $V_{it}^{\text{mkp-int}}$ and $V_{it}^{\text{mkp-non}}$, capture the contribution of intangible capital and non-intangible shocks to market power, respectively.

Because the non-intangible markup shock can be either positive or negative, its contri-

bution to firm value can also be negative relative to the zero-shock benchmark. A negative contribution indicates that the firm operates in a more competitive environment, which lowers its markup and firm value relative to the benchmark. Nevertheless, the total value of market power, $V_{it}^{\text{mkp}} = V_{it}^{\text{mkp-int}} + V_{it}^{\text{mkp-non}}$, is always positive.

4.4 Measure of the State Variables

To carry out the decomposition, I need to map all five state variables $(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it})$ for each firm-year onto the model's state space. Two capital inputs are directly observable in the data. I use Compustat item PPEGT for physical capital and the measure of intangible capital from Peters and Taylor (2017).⁵ I normalize total capital to match the model distribution:

$$\begin{aligned}\widehat{K+N}_{it} &= \overline{K+N} + \sigma^{K+N} \cdot z(K_{it} + N_{it}), \\ \hat{K}_{it} &= \widehat{K+N}_{it} \cdot \frac{K_{it}}{K_{it} + N_{it}}, \\ \hat{N}_{it} &= \widehat{K+N}_{it} \cdot \frac{N_{it}}{K_{it} + N_{it}},\end{aligned}\tag{19}$$

where $z(K_{it} + N_{it})$ is the cross-sectional z -score of firm i 's total capital in the data, and $\overline{K+N}$ and σ^{K+N} are the mean and standard deviation of total capital in the model simulation. This transformation removes the empirical time trend in the level while preserving the observed capital composition.

In the model, the markup is $\mu_{it} = (1 - \eta)P_{it}Y_{it}/P_{Lit}L_{it}$ under the assumption of Cobb-Douglas technology. Consistent with the model, I measure the empirical markup as $(1 - \eta) \cdot \text{SALE}_{it}/\text{COGS}_{it}$, where SALE and COGS are from Compustat, following De Loecker, Eeckhout and Unger (2020). I then back out the value of the exogenous markup shock $\hat{\zeta}_{it}^\mu$ using the estimated parameters and the markup function in Equation (2).

⁵Intangible capital may be measured with error because it is constructed from capitalized R&D and SG&A expenses, which may not fully capture assets such as brand loyalty and supplier relationships. Any resulting error is absorbed by productivity and markup shocks. Because capital and productivity shocks are grouped into the production component, its interpretation is largely unaffected. However, non-intangible market power may capture some effects of unmeasured intangibles. Section 4.6 discusses the interpretation of the two market power components.

Next, I estimate the two productivity terms z_{it}^K and z_{it}^N by minimizing the following objective function for each firm-year:

$$\begin{aligned} \min_{z^K, z^N} \quad & \left[\frac{V}{K+N}(z^K, z^N, \hat{\zeta}_{it}^\mu, \hat{K}_{it}, \hat{N}_{it}) - \frac{V_{it}}{K_{it+1} + N_{it+1}} \right]^2 / \sigma_{V/(K+N)}^2 \\ & + \left[\frac{N}{K+N}(z^K, z^N, \hat{\zeta}_{it}^\mu, \hat{K}_{it}, \hat{N}_{it}) - \frac{N_{it+1}}{K_{it+1} + N_{it+1}} \right]^2 / \sigma_{N/(K+N)}^2, \end{aligned} \quad (20)$$

where $\frac{V}{K+N}(\cdot)$ and $\frac{N}{K+N}(\cdot)$ are the model-implied ratios, and the errors are scaled by their variances $\sigma_{V/(K+N)}^2$ and $\sigma_{N/(K+N)}^2$ in the data.⁶ The system has two unknowns and two target moments, so in most firm-years the minimization achieves an exact match, though residuals remain for firms with extreme realizations. So, by construction, the model largely captures cross-sectional variation in valuations. The goal of the decomposition is to understand the composition of firm value rather than to explain its level.

With the estimated state variables $\hat{\mathbf{s}}_{it} \equiv (\hat{z}_{it}^K, \hat{z}_{it}^N, \hat{\zeta}_{it}^\mu, \hat{K}_{it}, \hat{N}_{it})$ for each firm-year, I decompose the firm value as:

$$V_{it} = V_{it}^{\text{model}} + V_{it}^\varepsilon \quad (21)$$

$$= V_{it}^{\text{phy}} + V_{it}^{\text{prod-int}} + V_{it}^{\text{mkp-int}} + V_{it}^{\text{mkp-non}} + V_{it}^\varepsilon, \quad (22)$$

where

$$\begin{aligned} V_{it}^{\text{model}} &\equiv (K_{it+1} + N_{it+1}) \cdot \frac{V}{K+N}(\hat{\mathbf{s}}_{it}), \\ V_{it}^j &\equiv V^j(\hat{\mathbf{s}}_{it}) \cdot \frac{K_{it+1} + N_{it+1}}{K(\hat{\mathbf{s}}_{it}) + N(\hat{\mathbf{s}}_{it})}, \end{aligned} \quad (23)$$

$$j \in \{\text{phy}, \text{prod-int}, \text{mkp-int}, \text{mkp-non}\},$$

and V_{it}^ε is actual firm value minus model-implied firm value.

⁶Since z^K and z^N contain both aggregate and firm-specific components, I use an iterative procedure that bounds each component using its model-implied unconditional volatility. Details are provided in Internet Appendix A.4.

4.5 Model Fit

Figure 3 plots the time series for the average estimated state variables as well as the standard deviation across firms in each year. Both z^K and z^N show fluctuations over the business cycle. Intangible productivity z^N has been steadily increasing over time and exhibits larger dispersion in the cross section, which suggests that firms differ more in the efficiency with which they utilize intangibles. Panels (c) and (d) plot the markup μ and the intangible capital share $N/(K + N)$, both of which trend upward over the sample. Moreover, firms become increasingly different in their market power, as indicated by the increasing cross-sectional dispersion of markups over time.

Figure 4(a) plots the cross-sectional R^2 in each year that measures how closely the model-implied firm value V_{it}^{model} matches the data counterpart V_{it} :

$$\hat{R}_t^2 = 1 - \frac{\sum_i (V_{it} - V_{it}^{\text{model}})^2}{\sum_i (V_{it} - \bar{V}_t)^2}, \quad (24)$$

where $\bar{V}_t$ is the cross-sectional mean of V_{it} . Unlike the standard regression R^2 , this measure does not allow for a free intercept or slope and therefore penalizes both level and scale deviations between the model and the data. On average, the R^2 is 87%, and it is stable over time and above 80% for most years, except for periods with special economic conditions such as the dot-com bubble and the 2008 financial crisis. I also compute R^2 for value scaled by capital, $V_{it}/(K_{it+1} + N_{it+1})$. The average R^2 for the value-to-capital ratio is 78%, and it remains stable over time as shown in Figure 4(b).

Direct evaluation of model fit at the firm level remains relatively uncommon in the literature. A closely related and important benchmark is Belo et al. (2022), who obtain a portfolio-level R^2 of 94% in matching model-implied and observed valuations in the cross section. Building on this line of work, I assess fit at the individual firm level and obtain an R^2 of around 80%. This comparison is inherently demanding, as firm-level analysis preserves idiosyncratic variation that is naturally attenuated through portfolio aggregation.

4.6 Validation

4.6.1 Example Firms

Figure 5 decomposes the average market value of representative firms into four model-implied components, computed at the firm-year level and averaged over time.

Technology and pharmaceutical firms derive substantial value from intangible capital, but through different channels. Amazon and Microsoft benefit from both production and market power, whereas Pfizer and Johnson & Johnson derive more of their intangible value through market power, consistent with the importance of patent protection and pricing power in pharmaceuticals. By contrast, firms such as Costco, 3M, and ExxonMobil derive more value from tangible capital.

The decomposition also reveals differences in non-intangible market power. Oracle, for example, has a sizable positive contribution, potentially reflecting network effects, switching costs, or other competitive advantages not captured by its current intangible stock. Costco and ExxonMobil instead exhibit negative contributions from this component, indicating that they face stronger competitive pressure, which pushes their markups below the levels implied by their intangible assets alone.

Finally, Amazon and Microsoft have substantial residual values, indicating that part of their market valuations is not explained by physical capital, intangible capital, or market power. This residual may reflect investor sentiment or other forces outside the model.

4.6.2 Value Share and Firm Characteristics

Figure 6(a) plots the relation between the intangible share and the total value of intangibles as a percentage of total value across firms. The intangible intensity is measured as the ratio of intangible capital to total capital, and the value of intangibles is the sum of the value of intangible capital in production and intangible-related market power. As expected, firms with higher intangible intensity tend to have a larger share of their value attributed to

intangibles. Similarly, Figure 6(b) plots the relation between the markup and the value of market power as a percentage of total value across firms. Firms with higher markups tend to derive more value from market power, and this relation is more concave.

4.6.3 Interpreting the Two Sources of Market Power

The two market power components capture distinct economic forces. They should respond to different types of shocks and vary with industry dynamics in different ways. In this section, I provide external validation for the decomposition by examining how the two market power components respond to innovation and competition shocks.

Patents and intangible market power. I first examine changes in the two market power value shares around the filing and issuance of highly cited patents, using the same event definition and citation threshold as in Section 2.1.3. I use citations rather than the economic value measure of Kogan et al. (2017) to avoid the mechanical effects as that measure is itself constructed from stock-market reactions, while the outcomes here are also value-based.

Figure 7 shows that the market power share rises around both filing and issuance, consistent with the markup response in Section 2.1.3. More importantly, the increase is concentrated in intangible-driven market power, while the non-intangible market power share remains broadly unchanged. This differential response supports the decomposition’s attribution of patent-related market power to intangible assets.

Tariff and non-intangible market power. Next, I exploit the granting of Permanent Normal Trade Relations (PNTR) status to China as a plausibly exogenous increase in product-market competition. Before PNTR, China’s access to low Normal Trade Relations tariffs required annual renewal, creating uncertainty because nonrenewal would trigger much higher non-NTR tariffs. PNTR permanently removed this uncertainty in 2001 and accelerated Chinese import competition. Following Pierce and Schott (2016), I measure industry exposure using the NTR gap, defined as the difference between non-NTR and NTR tariff

rates. I then estimate a difference-in-differences specification that interacts the industry-level NTR gap with a post-PNTR indicator.

Table 3 reports the responses of market power value shares to the PNTR shock. Panel A presents static difference-in-differences estimates, while Panel B traces the dynamic effects. Consistent with Corsini (2025), greater import competition does not significantly reduce the level of market power. Instead, it slows the growth of the market power share, with the effect driven entirely by non-intangible market power. Intangible-driven market power remains unchanged, consistent with this component capturing innovation-based rents rather than changes in the competitive environment.

Taken together, the patent and PNTR results provide external validation for the decomposition. Intangible-driven market power responds to innovation, whereas non-intangible market power is more sensitive to shifts in the external competitive environment.⁷

5 An Anatomy of Firm Value

This section presents the main quantitative results on firm value decomposition. Section 5.1 shows how the value of intangibles varies in the time series. Section 5.2 documents the composition shifts over firms' life cycles. Section 5.3 focuses on the cross-sectional variation.

5.1 Time Variation

Figure 8(a) shows the decomposition for the market, with value components summed across all firms. Figure 8(b) shows each component as a percentage of the total market valuation. Panel A of Table 4 formalizes the results and statistically tests the difference between the early and late periods. First of all, the value of intangible capital increases over the sample period. By 2024, it accounts for around 50% of the total market valuation explained by the

⁷Internet Appendix B.2 additionally shows that the two market power components exhibit different cross-sectional relationships with standard product-market characteristics, although these correlations should be interpreted descriptively because market structure and firms' intangible investments are jointly determined.

model, which amounts to around \$20 trillion in dollar terms.

The value of intangible capital in production grows steadily, increasing from 12.5% to 20.6% of total market value. On the other hand, the value of intangible-driven market power remains steady over time. Recent decades have seen substantial growth in technology, pushing up the production value of intangibles, whereas intangibles have long been a source of market power and so their contribution along this channel has remained relatively stable. Moreover, market power tends to be more valuable during economic downturns, because the ability to charge higher prices makes firms less vulnerable to negative shocks.

At the same time, the value attributed to non-intangible market power also grows. During the first half of the sample period, firms operate in a relatively competitive environment, with negative value from non-intangible sources. After 2000, however, firms increasingly earn rents in excess of intangible-driven markups, as reflected by the increasing non-intangible market power component. This rise in non-intangible market power provides suggestive evidence of a decline in competition, consistent with findings in the IO and macroeconomic literature (De Loecker, Eeckhout and Unger, 2020).

Finally, the residual component is generally small and becomes slightly more pronounced during market booms and busts. In the results that follow, I focus on the value components explained by the model and abstract from the residual term, which is not directly linked to the model's state variables and may reflect forces beyond the scope of this analysis.

Variation across time can be attributed to within-firm change or compositional change. For example, the value of market power can rise through either within-firm increases (all firms accumulate more market power) or compositional shifts (high-markup firms become larger). Following Paron (2025), I decompose the changes in the value components into these two channels. I use sales, $P_{it}Y_{it}$, to represent the firm-size distribution. Define firm i 's sales share as $w_{it} \equiv P_{it}Y_{it}/(P_tY_t)$ and its value-to-sales ratio as $v_{it} \equiv V_{it}/(P_{it}Y_{it})$. The aggregate value-to-sales ratio can then be written as a sales-weighted average of firm-level

value-to-sales ratios:

$$\frac{V_t}{P_t Y_t} = \sum_i w_{it} v_{it}. \quad (25)$$

The change in the value-to-sales ratio from $t - 1$ to t can be decomposed into within-firm and compositional change:⁸

$$\Delta \left(\frac{V_t}{P_t Y_t} \right) = \underbrace{\sum_i \bar{w}_{it} \Delta v_{it}}_{\text{Within-firm change}} + \underbrace{\sum_i \Delta w_{it} \bar{v}_{it}}_{\text{Compositional change}}. \quad (26)$$

For either $a = w$ or $a = v$, the overbar denotes the midpoint of the variable across the two periods:

$$\bar{a}_{it} \equiv \frac{1}{2} (a_{it} + a_{i,t-1}). \quad (27)$$

Intuitively, the within-firm change focuses on changes in firm-level valuations while holding sales weight fixed. The compositional change measures the effect of changing sales shares while holding valuations fixed for each firm. The total value V in Equation (26) can be replaced with each value component to understand their drivers.

Figure 9 plots the cumulative change in the aggregate value-to-sales ratio and decomposes it into within-firm and compositional components. It also shows the same decomposition for the ratio of the value components to sales. Consistent with Paron (2025), Panel (a) shows that the rise in aggregate valuation is driven mainly by compositional change, indicating a reallocation of revenue toward higher-valuation firms. Panel (b) shows that compositional changes contribute more than within-firm changes to the rise in the value of intangible capital in production, suggesting that firms have broadly increased intangible investment while those that use intangible capital more effectively have become more dominant.

Panels (c) and (d) isolate the contributions of intangible-driven and non-intangible mar-

⁸I account for entry and exit dynamics following Paron (2025). See their Appendix B for details.

ket power, respectively. Both market power components have increased over time relative to sales. While the rise in the value of non-intangible market power largely reflects compositional change, mirroring the pattern observed for total valuation, the increase in the value of intangible-driven market power is instead dominated by within-firm change. This contrast suggests that while many firms expand market power through the accumulation of intangible capital at the firm level, firms with less competition and higher pure rents are capturing a larger market share.

5.2 Firm Life Cycle

To understand the dynamics of firm value over the life cycle, I run the following regression and require firms included in the regression to be observed for at least 15 years to ensure a stable composition of firms in the sample:

$$\frac{V_{it}^{\text{comp}}}{V_{it}^{\text{model}}} = \sum_{j=1}^{26} \beta_j \cdot \mathbb{1}\{\text{age}_{it} = j\} + FE_t + FE_c + \varepsilon_{it}. \quad (28)$$

I define firm age as the number of years since the firm first appeared in CRSP, similar to Pástor and Pietro (2003) and Fama and French (2004). Following the literature on estimating life cycle effects (e.g., Fienberg and Mason, 1985), I include the year and cohort fixed effects in the regression. I group all observations at age 26 or older into a single age category. This restriction breaks the exact linear dependence among firm age, calendar year, and cohort, allowing the age indicators to be identified jointly with year and cohort fixed effects. I also report a specification that replaces the age dummies with a natural cubic spline in Internet Appendix B.3. The results are qualitatively similar.

Figure 10 traces how the value components evolve over firm age. Each panel plots the average component share against firm age, with the shaded area indicating 95% confidence intervals. Panel (a) shows that the physical capital component remains relatively stable over the firm life cycle, with a slight decline over the first five years.

Panel (b) shows a monotonic decline in the production value share of intangible capital of around 2%. One interpretation is that newly public firms enter the sample with relatively fresh and productive intangible stocks, such as recently developed technologies, novel organizational capital, or early-stage intellectual property. These productivity advantages erode over time as competitors imitate or as the intangibles themselves become obsolete. In contrast, Panel (c) documents that the value of intangible-driven market power rises by roughly 8%. This trajectory is consistent with firms progressively relying on their accumulated intangible capital to build and entrench market power, and the resulting rents become an increasingly important component of firm value.

Panel (d) shows that the non-intangible market power component declines over the firm life cycle. Because this component captures markup variation attributable to competitive conditions rather than the firm’s own intangible investments, the decline indicates that older firms tend to operate in increasingly competitive environments. This may reflect the entry of new competitors over time, or the fact that firms tend to enter the market during favorable periods with less competition. This pattern is directly in contrast with the intangible-driven market power component, which rises over the life cycle, suggesting that firms progressively rely more on intangible assets to preserve their market power.

Taken together, the four panels provide a coherent picture of firm value dynamics over the life cycle. Young firms benefit from fresh productive intangibles and operate in less competitive environments. As firms age, a compositional shift occurs within the intangible channel: the direct productivity value of intangibles decays, while firms increasingly convert their intangible stocks into durable market power.

5.3 Cross-sectional Heterogeneity

There is substantial heterogeneity in the composition of firm value across firms and across industries. I first perform a covariance decomposition to understand the contribution of value components to the cross-sectional variation. I scale both the actual and model-implied firm

value by total capital, $K_{it} + N_{it}$, so that the decomposition is not dominated by large firms. Let $v_{it} \equiv V_{it}/(K_{it} + N_{it})$ denote the scaled value (and analogously for each component). Equation (18) then implies:

$$\begin{aligned} Cov(v_{it}, v_{it}^{\text{model}}) &= Cov(v_{it}, v_{it}^{\text{phy}} + v_{it}^{\text{prod-int}} + v_{it}^{\text{mkp-int}} + v_{it}^{\text{mkp-non}}) \\ \Rightarrow 1 &= \frac{Cov(v_{it}, v_{it}^{\text{phy}})}{Cov(v_{it}, v_{it}^{\text{model}})} + \frac{Cov(v_{it}, v_{it}^{\text{prod-int}})}{Cov(v_{it}, v_{it}^{\text{model}})} + \frac{Cov(v_{it}, v_{it}^{\text{mkp-int}})}{Cov(v_{it}, v_{it}^{\text{model}})} + \frac{Cov(v_{it}, v_{it}^{\text{mkp-non}})}{Cov(v_{it}, v_{it}^{\text{model}})}. \end{aligned} \quad (29)$$

This equation decomposes the cross-sectional variation of the scaled firm value, v_{it} , that can be explained by the model, v_{it}^{model} . Figure 11 shows the results of the covariance decomposition for each cross section and Panel B of Table 4 reports the estimates. Consistent with the level decomposition, intangible capital plays a larger role in driving firm-level variation while physical capital becomes less important. Moreover, though the level of intangible-driven market power value is higher than the value of intangibles in production, the contribution of the latter to cross-sectional variation is much larger. This suggests that firms differ more in their production intangibles and their efficiency in utilizing intangibles in production. On the other hand, the total contribution of market power has increased substantially across time, accounting for roughly 29% of the value variation in the past two decades.

There is also substantial heterogeneity in value composition across sectors. Figure 12(a) shows the relative importance of intangible capital in production and market power across sectors. Sectors such as Business Equipment and Healthcare are located in the upper right corner, with a large portion of their value from both channels. On the other hand, sectors such as Manufacturing are located in the lower area, suggesting that the value of intangibles is mainly from raising production efficiency, while industries such as Telecom and Consumer Nondurables derive a large portion of value from intangible-driven market power.

The size and composition of market power also differ across sectors. Figure 12(b) plots the relative importance of intangible-driven and non-intangible market power across sectors. Sectors such as Healthcare and Business Equipment exhibit a large amount of non-intangible

market power, suggesting that consumer tastes or other exogenous shifts allow firms to charge higher prices. On the other hand, sectors including Chemicals and Consumer Durables mostly accumulate market power through intangibles, and they tend to face more intense competition in the market, which limits their ability to charge prices above the level justified by their intangible capital.

6 Applications

Distinguishing the production and market power roles of intangibles helps explain asset prices and corporate decisions. In this section, I focus on two applications using the value decomposition in the previous section. Section 6.1 creates model-implied value signals and improves the performance of the value premium in recent decades. Section 6.2 constructs Tobin's q using each value component and explores the investment- q relation.

6.1 Explaining the Value Premium Decline

A value signal, such as the book-to-market ratio, is designed to capture discount rate information through the ratio of firms' market value to fundamental value. If a firm is cheaper relative to its fundamental value, it will earn high returns going forward and vice versa. However, the value premium based on the traditional book-to-market ratio has declined over time, approaching zero after 2000 (Fama and French, 2021). Several papers have argued that book equity no longer captures the fundamental value accurately, so the book-to-market ratio becomes a noisy return signal. For example, Eisfeldt, Kim and Papanikolaou (2022) and Park (2022) propose to include intangible capital in the book equity. Gonçalves and Leonard (2023) instead replace the book equity with fundamental value by discounting future cash flows at a constant rate. However, the book-to-market ratio adjusted for intangibles still shows a visible decline in recent years. Although the fundamental-to-market ratio of Gonçalves and Leonard (2023) restores the value premium, the authors do not explain the

economic forces driving the wedge between book equity and fundamental equity.

I propose a different explanation. The book-to-market ratio has become less informative because market value increasingly reflects the value of market power, which is not captured by book equity. To fix ideas, consider a simple Gordon growth model with a constant growth rate and a firm-specific discount rate. Suppose the book equity reflects the value generated by production:

$$BE_i = \frac{CF_i^{\text{prod}}}{\bar{r} - g}, \quad (30)$$

while market equity contains both value from production and market power:

$$ME_i = \frac{CF_i^{\text{prod}} + CF_i^{\text{mkp}}}{r_i - g}. \quad (31)$$

Then the resulting book-to-market ratio is confounded by the share of cash flows attributable to production:

$$BM_i = \frac{BE_i}{ME_i} = \frac{r_i - g}{\bar{r} - g} \cdot \underbrace{\frac{CF_i^{\text{prod}}}{CF_i^{\text{prod}} + CF_i^{\text{mkp}}}}_{\text{Noise term}}. \quad (32)$$

6.1.1 Model-implied Proxies

To test this hypothesis, I construct two types of proxies. The first consists of model-implied valuation ratios that remove the value of market power from market value. I consider two variations: one makes a partial adjustment for market power and removes only the value of non-intangible markup from market value, while the other removes the total value of market power, including both intangible and non-intangible markups:

$$BM_i^{\text{model}} (\text{ex. exogenous markup}) = \frac{BE_i^{\text{total}}}{V_i^{\text{model}} - V_i^{\text{mkp-non}}}, \quad (33)$$

$$BM_i^{\text{model}} (\text{ex. all markup}) = \frac{BE_i^{\text{total}}}{V_i^{\text{model}} - V_i^{\text{mkp-non}} - V_i^{\text{mkp-int}}}, \quad (34)$$

where BE^{total} contains both the book equity from Fama and French (1992) and intangible capital, following the adjustment in Eisfeldt, Kim and Papanikolaou (2022).

These model-implied measures directly test whether heterogeneity in market power explains the decline in the value premium. However, they may be noisy because they rely on a model that abstracts from leverage and other potentially relevant factors. Therefore, I also construct a fitted fundamental-to-market ratio using the following regression:

$$\begin{aligned} \log FM_{i,t} = & \alpha_t + \beta_t \log \left(\frac{BE_{i,t}^{\text{total}}}{ME_{i,t}} \right) + \gamma_{1,t} \frac{V_{i,t}^{\text{prod-int}}}{V_{i,t}^{\text{model}}} \\ & + \gamma_{2,t} \frac{V_{i,t}^{\text{mkp-int}}}{V_{i,t}^{\text{model}}} + \gamma_{3,t} \frac{V_{i,t}^{\text{mkp-non}}}{V_{i,t}^{\text{model}}} + \varepsilon_{i,t}, \end{aligned} \quad (35)$$

where FM_{it} is the fundamental-to-market ratio proposed by Gonçalves and Leonard (2023). Each regressor is the value share of each component of the decomposition. I omit the value share of physical capital because it equals one minus the sum of the other three shares. I estimate the regression separately for each year t and use the fitted values, $\widehat{FM}_{i,t}$, to form portfolios for year $t+1$. Thus, portfolio sorts in the next section use only information available at the time of portfolio formation. Intuitively, the fitted fundamental-to-market ratio isolates the variation in the fundamental value that is captured by the model decomposition.

The regression coefficients in equation (35) shed light on the economic forces driving the wedge between FM and BM . Under the Gordon growth example in Equations (30) and (31), FE can be written as:

$$FE_i = \frac{CF_i^{\text{prod}} + CF_i^{\text{mkp}}}{\bar{r} - g}. \quad (36)$$

This expression suggests that variation in the wedge should be driven primarily by the market power components. Figure 13 plots the annual cross-sectional slope estimates and their 95% confidence intervals, while Table 5 reports estimates from a panel regression with year fixed effects.

Overall, the wedge between the fundamental-to-market and book-to-market ratios has

highly positive loadings on two market power terms, consistent with book equity not capturing the value of market power. The wedge loads negatively on the share of intangible production, which suggests that the book equity measure overestimates the production value. More importantly, the coefficients on both market power terms increase from the first to the second half of the sample, especially the coefficient on non-intangible market power. Meanwhile, R^2 is low when only $\log BE^{\text{total}}/ME$ is included in the post-2000 regression. When the three model-implied value shares are included, R^2 is close to the level in the first half of the sample.

Because the regression estimates the relationships flexibly rather than imposing the model’s exact quantitative predictions, the fitted ratio is less dependent on specific model assumptions. It therefore isolates the component of fundamental value associated with the model-implied channels, especially market power. If the fitted ratio restores the value premium to a similar extent as the original fundamental-to-market ratio, the result provides further evidence that the value premium decline is driven primarily by the growing importance of market power.

6.1.2 The Value Premium Decline

I first look at the value premium in the full sample using different value signals. Panel A of Table 6 reports the return spreads from decile portfolio sorts from July 1977 to June 2026. The two book-to-market ratios constructed directly from the data generate monthly return spreads of 0.19% and 0.33%, neither of which is statistically significant. The model-implied ratio that removes only non-intangible market power also produces a small and insignificant spread of 0.13% ($t = 0.82$). In contrast, removing the value of both intangible and non-intangible market power yields a significant value premium of 0.33% per month ($t = 2.12$). The fitted fundamental-to-market ratio produces the largest spread, at 0.49% per month ($t = 3.13$), while the original fundamental-to-market ratio generates a spread of 0.45% ($t = 2.30$). Its stronger performance suggests that the model-implied value shares

capture the information in the original fundamental-to-market ratio that is most relevant for predicting returns.

Next, I examine the time series of the value premium to assess whether it has declined over time. Figures 14a and 14b plot its ten-year and five-year trailing averages, respectively. The figures compare the intangible-adjusted book-to-market ratio with FM , the model-implied ratio that removes the value of all market power, and fitted FM . The data benchmark exhibits the clearest decline: its trailing average falls sharply after the mid-2000s and becomes negative in recent years. The value premiums based on FM and the model-implied ratio also weaken and turn negative during parts of the recent period, although their declines are less pronounced. By contrast, fitted FM does not exhibit a persistent downward trend. Its ten-year trailing average remains mostly positive, and its five-year trailing average rebounds strongly after a temporary decline in the early 2020s.

To formally test whether the value premium has declined, I run a simple time trend regression on the monthly value premium series.⁹ For each monthly return spread r_t I estimate

$$r_t = a + b \cdot \frac{t}{T} + \varepsilon_t, \quad t = 1, \dots, T, \quad (37)$$

where b is the total drift across the sample in % per month. Panel B of Table 6 reports the results. Four of the measures exhibit a declining trend, with estimated b significantly negative. In contrast, fitted FM shows no significant decline, with $b = -0.34$ and a t -statistic of -0.54 . The model-implied ratio that removes all market power shows the second-best performance across measures, with $b = -0.93$. Although this estimate is significant at the 10% level because it is estimated more precisely, its magnitude is smaller than those of the remaining value signals. This finding suggests that removing the value of market power attenuates the downward trend in the value premium, while the flexible adjustment embodied

⁹I do not assess the decline in the value premium using subsample comparisons because the results are sensitive to the choice of the sample split date. Internet Appendix B.4 illustrates this sensitivity by reporting value premium estimates for three alternative split dates. Nevertheless, the fitted fundamental-to-market ratio delivers the largest value premium in the second subsample across all specifications.

in fitted FM attenuates it further. Together with the decomposition results, this pattern is consistent with the growing importance of market power contributing to the decline of conventional value signals.

6.2 The Investment- q Relation

The neoclassical theory of investment predicts that Tobin’s q , the ratio of capital’s market value to its replacement cost, should fully capture investment opportunities (Hayashi, 1982). As firms increasingly rely on intangible assets, Peters and Taylor (2017) further incorporate intangible capital into the replacement cost and show that this total q measure better predicts investment. However, a large total q may reflect the market power rents, which raise the firm value without increasing the marginal return to capital (Gutiérrez and Philippon, 2017; Crouzet and Eberly, 2019). More importantly, firms may have very different investment incentives even when they share the same total q , because that market value may arise from fundamentally different economic sources. My decomposition framework provides market-power-adjusted and component-specific q measures and thus helps explain the different incentives for investments.

6.2.1 Summary statistics

Panel A of Table 7 reports summary statistics for the empirical and model-implied q measures. Standard q , defined as the market value of the firm relative to physical capital, has a mean of 4.73 and exhibits substantial dispersion and positive skewness. Incorporating intangible capital into the capital stock substantially lowers the level of total q , with the mean falling to 1.42, while maintaining a similar distribution.¹⁰ The next three measures are counterparts to the data total q , where I purge out the value of residuals, non-intangible

¹⁰Note that the definition of total q differs slightly from that in Peters and Taylor (2017), where they subtract current assets (Compustat item ACT) from firm value. Here I subtract cash (CHE) to be consistent with my definition of firm value throughout the paper. Internet Appendix B.5 reports the regressions in the next two subsections for the exact Peters-Taylor q . The results are similar to those obtained using this paper’s total q measure.

market power, or the total value of market power. They have similar distributions but different levels. The component q measures show that physical and intangible capital contribute comparably to firm value on average, with mean physical and intangible q of 1.47 and 1.43, respectively. Within intangible capital, the market power component (0.84) exceeds the production component (0.59), indicating that a substantial share of intangible value reflects firms' ability to generate rents rather than improvements in production efficiency. Overall, the level and distribution of model-implied q measures are reasonable and comparable to the data benchmarks.

Panel B of Table 7 reports the correlation matrix among these q measures. Model total q 's covary with the data counterpart with a correlation of around 0.8. At the same time, the decomposed q measures are far from perfectly correlated. In particular, the production and market power components of intangible q have a correlation of only 0.40, and physical q is only weakly correlated with intangible market power q (0.08). These moderate correlations suggest that the decomposition isolates economically distinct sources of firm value rather than mechanically partitioning total q .

6.2.2 Horse race

Table 8 reports the horse race of model-implied total q 's against the data benchmarks. Panel A regresses physical investment onto q measures and Panel B uses intangible investment as the left-hand variable.¹¹ The first column reports the investment- q relation using standard Tobin's q . It explains some variation in both investment rates with very small coefficients. Under the neoclassical framework, the inverse q slope measures the adjustment costs. There-

¹¹Here I scale each investment rate by its own capital rather than by total capital. This is the main specification for two reasons. First, the model specifies separate adjustment-cost functions for physical and intangible capital, so each first-order condition is expressed in terms of its corresponding investment rate. Second, scaling by total capital mechanically introduces variation in capital composition. For example, even in steady state, a firm with a larger intangible share will have lower physical investment relative to total capital. Because intangible component q covaries with the intangible share, this mechanical relation would bias its estimated coefficient downward for physical investment- q regressions. Nevertheless, Internet Appendix B.5 reports the results for $(K+N)$ -scaled investment rates. They are qualitatively similar to the main results.

fore, the small coefficient implies an unreasonably large adjustment cost. Data total q in the second column predicts investment with a much larger R^2 and a more reasonable slope.

If market power value and residual valuation components do not generate the same investment incentives as productive capital, removing them from total q should increase the estimated investment- q slope. Columns (3)–(5) of Table 8 examine this prediction. The adjusted model-implied total q measures deliver within- R^2 values comparable to those of the empirical benchmarks, but their coefficients are roughly twice as large. Under the standard quadratic adjustment-cost interpretation, these larger coefficients imply more moderate adjustment costs. The coefficients are smaller for intangible investment than for physical investment, consistent with greater adjustment frictions for intangible capital.¹²

6.2.3 Component q

Panel A of Table 9 reports investment- q regressions using the individual value components. Columns (1)–(3) examine physical investment. Physical q alone explains more than 9% of the within-firm variation in physical investment. When all four components are included in column (2), physical q remains the dominant predictor, while the intangible production and market power components have substantially smaller coefficients. Column (3) adds total q to the component regression. The coefficient on non-intangible market power q becomes significantly negative, suggesting that total q overstates investment incentives when it assigns the same role to valuation arising from productive capital and from non-intangible market power. By contrast, physical q and intangible market power q remain positively related to physical investment, indicating that total q may underweight these sources of value.

Columns (4)–(6) report the corresponding results for intangible investment. Firms may invest in intangible capital either to improve production efficiency or to build market power. Consistent with this interpretation, both intangible-production q and intangible market

¹²Table 1 reports a lower adjustment cost parameter for intangible capital. However, I do not allow disinvestment in intangible capital, so the overall adjustment friction is higher for intangible capital, consistent with the smaller investment- q slope.

power q strongly predict future intangible investment in columns (4) and (5), with the market power component carrying the larger coefficient. In column (6), intangible market power q remains strongly positive even after controlling for total q , whereas intangible-production q becomes negative. This pattern suggests that the market power channel is an important determinant of intangible investment beyond the information contained in aggregate valuation. Non-intangible market power q , in contrast, is economically small.

To quantify the relative importance of the components, Panel B of Table 9 decomposes the regression R^2 using Shapley values. Variation in physical investment is explained primarily by physical q , whereas variation in intangible investment is concentrated in the two intangible components, especially intangible market power q . Non-intangible market power q contributes little to the explanatory power of either regression.

Taken together, the results show that aggregate q obscures economically distinct investment incentives. Production components predict investment in the corresponding capital stock, while intangible-driven market power value provides an additional and particularly important incentive for intangible investment. Non-intangible market power value, by contrast, raises firm valuation without generating comparable investment incentives. This distinction helps explain why conventional total q can deliver attenuated investment slopes even when its overall predictive power remains similar.

7 Conclusion

Firms actively acquire market power through intangible investments, such as marketing and product differentiation. This paper extends the neoclassical framework to incorporate monopolistic competition and the endogenous link between intangibles and market power. In the model, firms invest in intangible capital to both raise productivity and build market power. I estimate the model and provide a quantitative framework to decompose the intangible value into production and market power components. I show that, on average,

intangibles generate more value through market power than through production.

Over time, intangible capital and market power account for a growing share of total market valuation, with market power increasingly tilting toward non-intangible sources. The rise in intangible-driven market power is primarily a within-firm shift, while the rise in non-intangible market power is largely driven by reallocation toward high-valuation firms. Over the firm life cycle, young firms enter with productive intangible stocks and favorable competitive conditions, but the direct productivity value of intangibles decays with age while firms increasingly convert their intangible capital into durable market power.

I show that incorporating the value of intangible market power is important for understanding asset prices and incentives for intangible investments. Specifically, the value of market power could contaminate the book-to-market ratio, contributing to the value premium decline in recent decades. Removing the value of market power explains part of the decline. Moreover, I construct a separate q measure for each economic component and show that firms' intangible investment reacts strongly to market power incentives. In contrast, non-intangible market power value raises firm valuation without generating comparable investment incentives.

This framework offers several avenues for future research. First, richer data could be used to examine how different types of intangibles contribute to market power through distinct mechanisms. Second, although this paper focuses on the composition of firm value and the dual roles of intangibles, the framework can be adapted to study other questions in finance. For example, differences in the risk exposures of value components may help explain cross-sectional variation in expected returns. Finally, extending the analysis beyond U.S. firms could reveal whether market power has also increased in other countries and whether its rise in the United States has come at the expense of foreign firms.

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Table 1: Parameter estimates

This table reports the model parameters. Panel A reports the calibrated parameters. Panel B lists the estimated parameters and their standard errors. The model is solved and simulated at annual frequency.

Variable	Notation	Value	SE
Panel A: Calibrated parameters			
Discount rate	r_f	0.04	
Scaling of profits	$\bar{Y}$	20	
Depreciation of physical capital	δ^K	0.12	
Depreciation of intangible capital	δ^N	0.2	
Panel B: Estimated parameters			
<i>Group 1: Production technology</i>			
Share of physical capital	α	0.7297	0.0916
Total share of capital	η	0.2662	0.0048
CES curvature	ρ	0.4515	0.0619
Fixed operating cost	f	0.3766	0.0274
<i>Group 2: Investment and capital-specific productivity</i>			
Adjustment cost of physical capital	b^K	2.9059	0.1957
Adjustment cost of intangible capital	b^N	1.9468	0.3501
Autocorrelation of z_t^K	ρ^K	0.8131	0.0342
Volatility of ζ_t^K shocks	σ_ζ^K	0.6106	0.1521
Autocorrelation of z_t^N	ρ^N	0.9788	0.0126
Volatility of ζ_t^N shocks	σ_ζ^N	0.2227	0.0825
<i>Group 3: Markup-related</i>			
Average effect of intangible capital on markup	$\bar{\mu}$	0.1267	0.0134
Curvature of the markup function	ϕ	0.7576	0.0259
Autocorrelation of ζ_t^μ	ρ^μ	0.7587	0.0151
Volatility of ζ_t^μ shocks	σ^μ	0.1456	0.0036
<i>Group 4: SDF</i>			
Volatility of ε_x^K	σ_x^K	0.0437	0.0153
Volatility of ε_x^N	σ_x^N	0.2344	0.0112
Risk price of ε_x^K	γ^K	19.9670	14.8757
Risk price of ε_x^N	γ^N	16.3640	2.8020

Table 2: Moments

This table reports moments used in the SMM. The model is solved and simulated at an annual frequency. I simulate the model for 100 independent samples, with each sample including 3500 firms over 200 years (the first 150 years form a burn-in period). I report the data moments and the means of the corresponding simulated moments. The SE column reports the sampling standard error of each empirical moment, calculated from the influence-function covariance matrix. Firm-level moments account for clustering by firm, while the aggregate moments treat calendar year as the unit of observation. Definitions of the variables are in Table [IA.1](#).

Moment	Data	Simulated	SE
Panel A: Firm-level moments			
Average value-to-total-capital ratio	1.3834	1.4425	0.0133
Std. dev. of physical investment	0.1213	0.1213	0.0009
Std. dev. of intangible investment	0.1189	0.1170	0.0010
Average intangible share	0.4839	0.4917	0.0033
Std. dev. of intangible share	0.0852	0.0818	0.0011
Average profits-to-total-capital ratio	0.4284	0.4350	0.0027
AR(1) of profits-to-total-capital ratio	0.6813	0.6424	0.0047
Std. dev. of profits-to-physical-capital ratio	0.6947	0.7376	0.0101
AR(1) of profits-to-physical-capital ratio	0.6935	0.6969	0.0073
Std. dev. of profits-to-intangible-capital ratio	0.9142	0.7794	0.0152
AR(1) of profits-to-intangible-capital ratio	0.6397	0.6260	0.0090
Average SALE/COGS	1.7538	1.7963	0.0097
Std. dev. of SALE/COGS	0.3960	0.3749	0.0082
AR(1) of SALE/COGS	0.6433	0.7532	0.0130
Cross-sectional correlation between SALE/COGS and intangible share	0.2487	0.3431	0.0093
Panel B: Aggregate moments			
Average market excess return	0.0893	0.0753	0.0240
Average HML sorted by profitability	0.0402	0.0415	0.0276
Volatility of aggregate profits-to-total-capital	0.0361	0.0336	0.0030
AR(1) of aggregate profits-to-total-capital	0.8681	0.8320	0.0085
Volatility of aggregate physical investment	0.0218	0.0064	0.0024
Volatility of aggregate intangible investment	0.0152	0.0173	0.0013

Table 3: PNTR exposure and market-power value shares (1991–2007)

This table reports difference-in-differences estimates of the effect of PNTR-driven import competition on the two market-power value shares; Section 4.6.3 describes the setting and the NTR gap measure of industry exposure. The Level columns use the value share itself, and the Growth columns use its annual change. Panel A interacts the NTR gap with a post-2000 indicator. Panel B uses separate interactions for the distant pre-period, the short run, and the medium run, omitting 1997–2000 as the reference period. All specifications include firm and year fixed effects. Standard errors, in parentheses, are clustered by NTR-gap industry family, the level at which Pierce and Schott (2016) define the tariff gap.

	Intangible market-power share		Non-intangible market-power share	
	Level	Growth	Level	Growth
Panel A: Static difference-in-differences				
Gap $\times$ Post	0.0304 (0.0193)	0.0064 (0.0060)	0.0048 (0.0446)	-0.0209*** (0.0061)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	52,505	44,245	52,505	44,245
R^2	0.7418	0.2230	0.7841	0.1411
Within R^2	0.00081	0.00005	0.00001	0.00023
Panel B: Dynamic difference-in-differences				
Gap $\times$ Pre (1991–1996)	0.0010 (0.0139)	-0.0093 (0.0071)	0.0011 (0.0181)	-0.0112 (0.0077)
Gap $\times$ Short run (2001–2003)	0.0264 (0.0161)	0.0032 (0.0083)	0.0165 (0.0416)	-0.0270*** (0.0099)
Gap $\times$ Medium run (2004–2007)	0.0355 (0.0236)	0.0025 (0.0072)	-0.0069 (0.0435)	-0.0234*** (0.0089)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	52,505	44,245	52,505	44,245
R^2	0.7418	0.2230	0.7841	0.1412
Within R^2	0.00085	0.00013	0.00010	0.00028

Table 4: Aggregate decomposition

This table reports the level and covariance decomposition of V into terms attributed to physical capital, intangibles in production, intangible-driven market power, and non-intangible market power. Panel A presents the percentage contribution of each term to total market value. Each term is calculated as $\sum_i V_{it}^{\text{comp}} / \sum_i V_{it}^{\text{model}}$ and then averaged over time. Panel B reports the covariance decomposition following Equation (29) (average across years). The Late – Early row reports the difference between the late and early subperiod averages; standard errors are in parentheses.

		% V^{phy}	% $V^{\text{prod-int}}$	% $V^{\text{mkp-int}}$	% $V^{\text{mkp-non}}$
Panel A: Level decomposition					
Full sample	(1976–2024)	55.11	16.47	28.33	0.10
Early	(1976–2000)	63.74	12.56	27.74	-4.05
Late	(2001–2024)	46.11	20.54	28.93	4.42
Late – Early		-17.64	7.98	1.19	8.47
		(1.15)	(1.12)	(0.75)	(0.80)
Panel B: Covariance decomposition					
Full sample	(1976–2024)	53.04	22.65	13.01	11.30
Early	(1976–2000)	63.72	16.49	10.79	9.00
Late	(2001–2024)	41.91	29.06	15.32	13.71
Late – Early		-21.81	12.57	4.53	4.71
		(1.85)	(1.81)	(0.45)	(0.81)

Table 5: The fitted value signal, book-to-market, and market-power value shares

This table regresses the log fundamental-to-market ratio (FE/ME) on the log book-to-market ratio and the model-implied value shares of intangible production, intangible markup, and non-intangible markup. All specifications include year fixed effects. Standard errors, two-way clustered by firm and year, are in parentheses.

	Full sample		1976–1999		2000–2024	
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(BE^{\text{total}}/ME)$	0.0964*** (0.0277)	0.0762*** (0.0277)	0.2269*** (0.0192)	0.1829*** (0.0345)	-0.0346 (0.0361)	-0.0415 (0.0319)
Intangible (production)		-0.4284*** (0.0359)		-0.4947*** (0.0407)		-0.3818*** (0.0525)
Intangible (markup)		0.7558*** (0.1023)		0.6387*** (0.1509)		0.8798*** (0.1169)
Non-intangible markup		0.8616*** (0.0542)		0.6237*** (0.0842)		1.0171*** (0.0475)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0696	0.1982	0.1195	0.2021	0.0268	0.1995
Within R^2	0.0086	0.1457	0.0573	0.1458	0.0010	0.1782
Observations	102,518	102,518	54,155	54,155	48,363	48,363

Table 6: Value premiums and trend regressions

Panel A reports the value premium, value minus growth in % per month, from yearly NYSE-breakpoint decile sorts, value weighted by lagged market equity, from July 1977 to June 2026 (588 months). Panel B reports estimates of $r_t = a + b(t/T) + e_t$ for the corresponding monthly value-premium spreads. The intercept a is the fitted premium at the start of the sample, b is the total drift across the sample, both in % per month, and b/σ expresses the drift in standard deviations of the series. Newey–West t -statistics using automatic lag selection are in parentheses beneath the Panel A estimates and the a and b estimates in Panel B. For the BM ratios, the value portfolio is the low market-to-book decile; for fitted FM and FM , it is the high decile.

Sorting variable	Panel A: Value premium	Panel B: Trend regressions		
	% monthly	a	b	b/σ
<i>Data ratios</i>				
BM^{data} (original)	0.19 (0.72)	0.83 (1.82)	-1.27 (-1.32)	-0.2290
BM^{data} (incl. intangibles)	0.33 (1.27)	1.19 (2.60)	-1.72 (-2.05)	-0.3311
<i>Model-implied ratios</i>				
BM^{model} (ex. exogenous markup)	0.13 (0.82)	0.81 (2.50)	-1.35 (-2.70)	-0.3463
BM^{model} (ex. all markup)	0.33 (2.12)	0.79 (2.53)	-0.93 (-1.67)	-0.2609
<i>Fundamental-to-market</i>				
Fitted FM	0.49 (3.13)	0.66 (1.98)	-0.34 (-0.54)	-0.0885
FM	0.45 (2.30)	1.06 (3.02)	-1.23 (-1.88)	-0.2667

Table 7: Properties of the q measures

This table reports the properties of the q measures used in the investment regressions: Panel A gives summary statistics and Panel B the pooled correlation matrix. Every measure divides a value by end-of-period capital, where K is the physical-capital stock and N is the intangible-capital stock; V is the sum of market equity and the book value of debt, minus cash and short-term investments. $V^{\text{model}} = V^{\text{phy}} + V^{\text{prod-int}} + V^{\text{mkp-int}} + V^{\text{mkp-non}}$ is the model-implied value, which removes the unexplained residual from market value. Skew is the sample skewness $g_1 = m_3/m_2^{3/2}$, which is zero under symmetry. All q measures are winsorized within each year at the 1st and 99th percentiles, and the sample runs from 1976 to 2024.

Panel A: Summary statistics

Measure	Definition	Mean	SD	Median	Skew
Standard q	V/K	4.73	8.73	1.98	5.46
Total q	$V/(K+N)$	1.42	1.77	0.92	5.36
Model total q	$V^{\text{model}}/(K+N)$	1.33	0.98	1.06	1.69
ex. non-intangible markup	$(V^{\text{model}} - V^{\text{mkp-non}})/(K+N)$	1.28	0.87	0.98	1.54
ex. all markup	$(V^{\text{model}} - V^{\text{mkp-int}} - V^{\text{mkp-non}})/(K+N)$	0.95	0.78	0.70	1.51
Physical q	V^{phy}/K	1.47	1.72	0.87	3.30
Intangible q (total)	$(V^{\text{prod-int}} + V^{\text{mkp-int}})/N$	1.43	1.03	1.13	3.10
Intangible q (production)	$V^{\text{prod-int}}/N$	0.59	0.61	0.37	2.10
Intangible q (markup)	$V^{\text{mkp-int}}/N$	0.84	0.62	0.71	4.19

Panel B: Pooled correlation matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Standard q	1.00								
(2) Total q	0.74	1.00							
(3) Model total q	0.62	0.82	1.00						
(4) ex. non-intangible markup	0.58	0.81	0.91	1.00					
(5) ex. all markup	0.49	0.79	0.86	0.98	1.00				
(6) Physical q	0.70	0.73	0.76	0.83	0.81	1.00			
(7) Intangible q (total)	0.25	0.45	0.54	0.52	0.55	0.33	1.00		
(8) Intangible q (production)	0.41	0.58	0.63	0.71	0.70	0.48	0.83	1.00	
(9) Intangible q (markup)	0.01	0.18	0.28	0.16	0.23	0.08	0.84	0.40	1.00

Table 8: Investment and q : horse race

This table reports the investment- q regressions for different total q measures. Each column is a univariate regression of the investment rate on a single q . Panel A uses physical investment I^K/K and Panel B intangible investment I^N/N . Both investment rates are measured one year ahead. Model total q is the model-implied value over $K+N$; the last two columns strip the non-intangible market-power component $V^{\text{mkp-non}}$, and then both market-power components $V^{\text{mkp-int}}$ and $V^{\text{mkp-non}}$, out of that value. All q measures and investment rates are winsorized within each year at the 1st and 99th percentiles. All specifications include firm and year fixed effects, with standard errors two-way clustered by firm and year in parentheses.

	Standard q	Total q	Model total q	Model total q (ex. non-intangible markup)	Model total q (ex. all markup)
	(1)	(2)	(3)	(4)	(5)
Panel A: Physical investment, I^K/K					
q	0.0067*** (0.0005)	0.0327*** (0.0030)	0.0662*** (0.0034)	0.0691*** (0.0035)	0.0736*** (0.0036)
Observations	105,979	105,979	105,979	105,979	105,979
R^2	0.4546	0.4748	0.4741	0.4718	0.4681
Within R^2	0.0934	0.1271	0.1258	0.1220	0.1159
Panel B: Intangible investment, I^N/N					
q	0.0039*** (0.0004)	0.0292*** (0.0023)	0.0571*** (0.0031)	0.0571*** (0.0030)	0.0649*** (0.0033)
Observations	105,979	105,979	105,979	105,979	105,979
R^2	0.5899	0.6308	0.6261	0.6201	0.6242
Within R^2	0.0446	0.1398	0.1288	0.1150	0.1245

Table 9: Investment and q : value components

Panel A reports investment- q regressions using the value components. Columns (1)–(3) explain physical investment I^K/K and columns (4)–(6) intangible investment I^N/N ; both dependent variables are measured one year ahead. Components are scaled by K in columns (1)–(3) and by N in columns (4)–(6). Columns (1) and (4) use the components generated by the corresponding capital stock, columns (2) and (5) use all four components, and columns (3) and (6) add total q . Panel B decomposes the within- R^2 of columns (2) and (5) into Shapley shares, which add to 100. All q measures and investment rates are winsorized within year at the 1st and 99th percentiles. All specifications include firm and year fixed effects; standard errors in Panel A are two-way clustered by firm and year.

	Physical investment, I^K/K			Intangible investment, I^N/N		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Investment regressions						
V^{phy}	0.0291*** (0.0025)	0.0268*** (0.0024)	0.0092*** (0.0023)		0.0011** (0.0005)	0.0010** (0.0004)
$V^{\text{prod-int}}$		0.0049*** (0.0012)	-0.0025** (0.0011)	0.0255*** (0.0026)	0.0274*** (0.0027)	-0.0072** (0.0031)
$V^{\text{mkp-int}}$		0.0041* (0.0022)	0.0201*** (0.0025)	0.0599*** (0.0063)	0.0442*** (0.0072)	0.0353*** (0.0066)
$V^{\text{mkp-non}}$		0.0040*** (0.0013)	-0.0044*** (0.0012)		0.0004 (0.0006)	0.0012** (0.0005)
Total q			0.0264*** (0.0031)			0.0272*** (0.0023)
Observations	105,979	105,979	105,979	105,979	105,979	105,979
R^2	0.4539	0.4592	0.4862	0.6022	0.6028	0.6420
Within R^2	0.0923	0.1011	0.1459	0.0732	0.0747	0.1659
Panel B: Shapley decomposition of within- R^2 (%)						
	Physical investment, I^K/K			Intangible investment, I^N/N		
V^{phy}		76.72			21.50	
$V^{\text{prod-int}}$		13.09			32.14	
$V^{\text{mkp-int}}$		9.00			40.83	
$V^{\text{mkp-non}}$		1.20			5.53	
Total within R^2		0.1011			0.0747	
Observations		105,979			105,979	

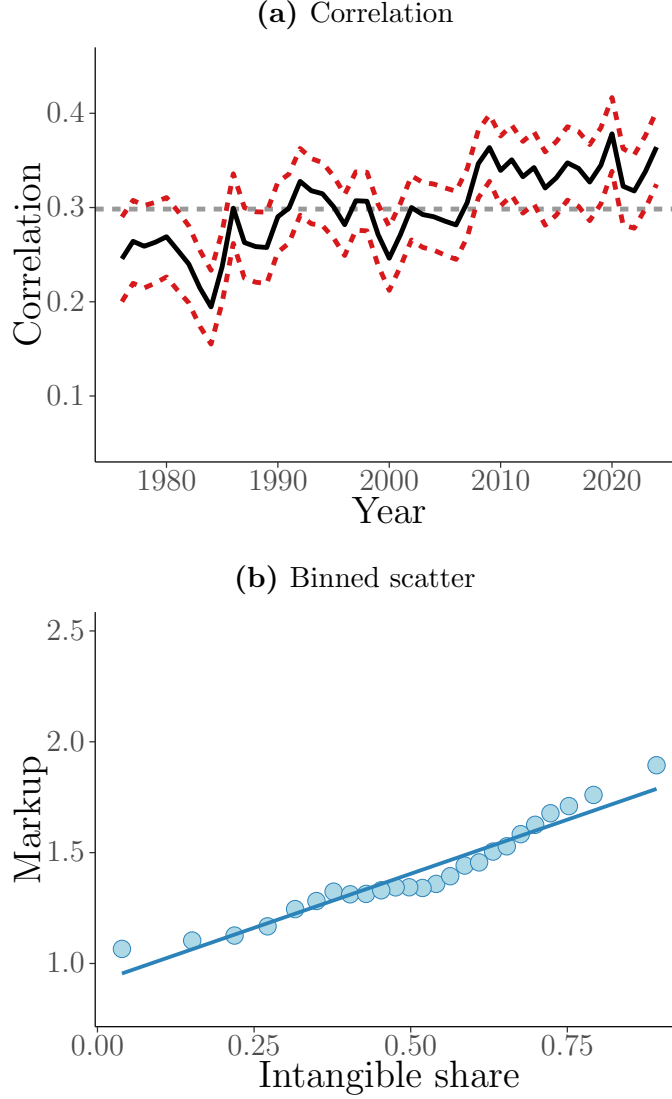


Figure 1: Cross-sectional correlation between intangible share and markups (1976–2024)

This figure plots the relationship between intangible capital share and markups, using the Cobb-Douglas estimate that includes intangible capital as a production input. Panel (a) plots the annual cross-sectional correlation, demeaning both variables within each two-digit industry-year cell, with the 95% confidence interval in red and the sample average in gray. Panel (b) plots a binned scatter: both variables are residualized on two-digit industry and year fixed effects, sorted into 25 equal-sized bins on the residualized intangible share, and plotted as bin means with the sample means added back, with an OLS fitted line. Markups are winsorized within year at the 2.5th and 97.5th percentiles. Intangible share is intangible capital divided by the sum of physical and intangible capital, with intangible capital from Peters and Taylor (2017) and physical capital measured by Compustat item PPEGT.

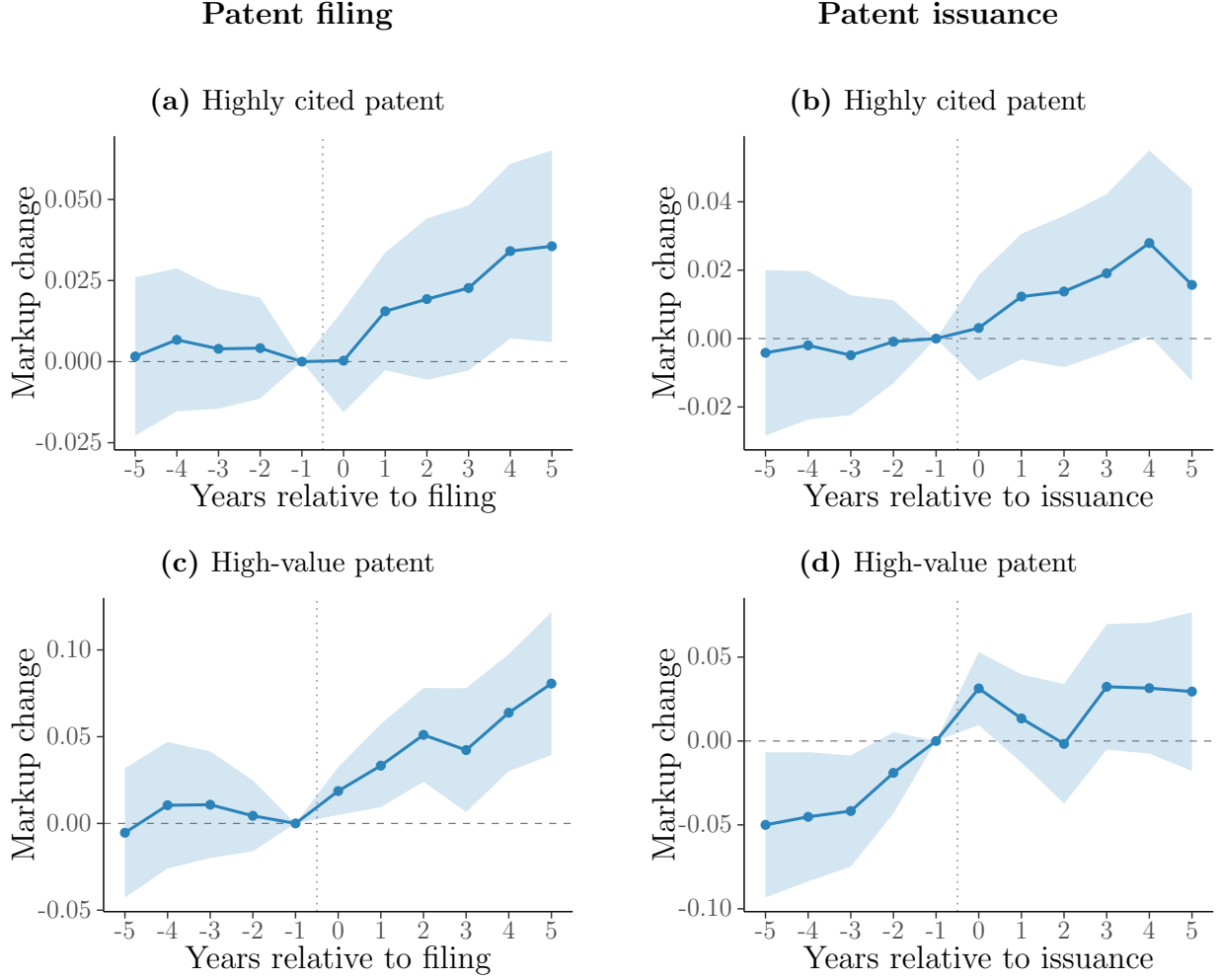


Figure 2: Event studies under Cobb-Douglas with intangibles

This figure plots dynamic treatment effects of important patent filings (left column) and issuances (right column) on markups, using the Cobb-Douglas production-function estimate that includes intangible capital as a production input. Panels (a) and (b) classify a patent as important if its forward-citation count lies in the top 10% within its Fama-French 48 industry and filing or issuance year; Panels (c) and (d) instead classify a patent as important if its stock-market value, as measured by Kogan et al. (2017), lies in the top 10% within the same industry-year cell. A firm is treated in the year of its first important patent, and firms that never have an important patent serve as the never-treated control group. Coefficients are estimated with the Sun and Abraham (2021) interaction-weighted estimator and are normalized relative to the year before the event ($\tau = -1$). The specification includes firm and year fixed effects. The estimation window is $\tau \in [-5, +5]$. Shaded areas are 95% confidence intervals, with standard errors clustered by firm.

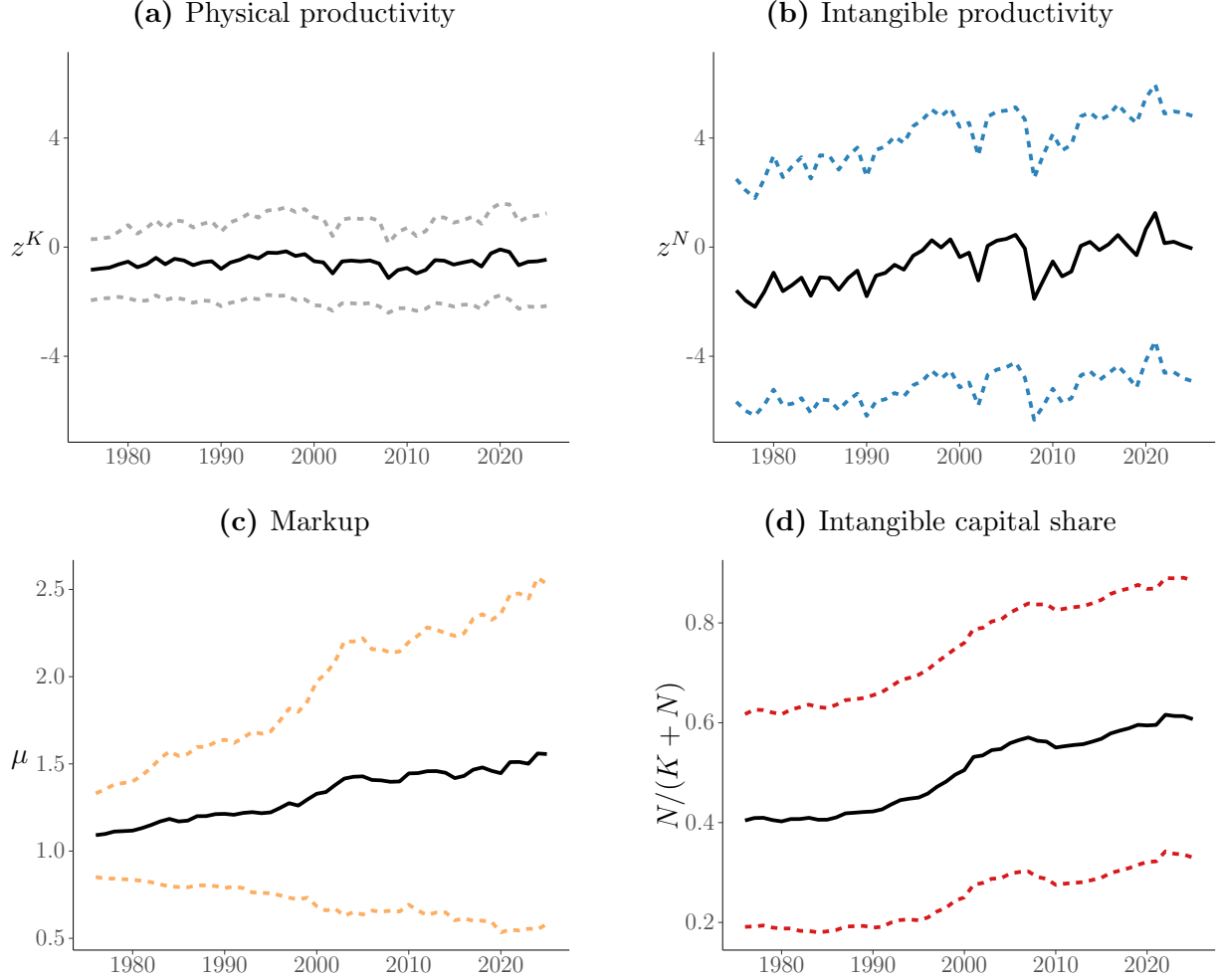


Figure 3: Time series of estimated state variables (1976–2024)

This figure plots the estimated state variables, averaged over firms in each year. Panels (a) and (b) show the cross-sectional average productivity for physical capital, $\bar{z}_t^K = x_t^K$, and intangible capital, $\bar{z}_t^N = x_t^N$, respectively. Panels (c) and (d) show the cross-sectional average markup, $\bar{\mu}_t$, and intangible capital share, $\bar{N}/(\bar{K} + \bar{N})_t$, respectively. In each panel, the solid line is the cross-sectional mean and the dashed lines represent the mean $\pm$ one cross-sectional standard deviation, reflecting the dispersion across firms.

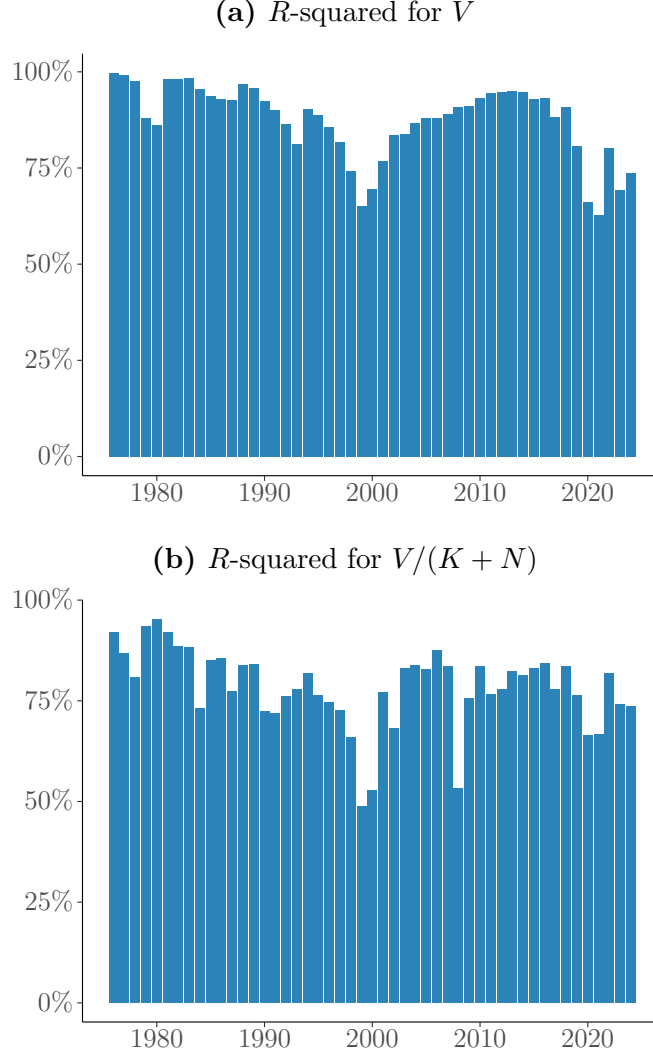


Figure 4: Cross-sectional R -squared over time (1976–2024)

This figure plots the cross-sectional R -squared defined as $\hat{R}_t^2 = 1 - \frac{\sum_i (V_{it} - V_{it}^{\text{model}})^2}{\sum_i (V_{it} - \bar{V}_t)^2}$, which restricts both $a = 0$ and $b = 1$. Panel (a) uses firm value V_{it} and panel (b) uses V_{it} scaled by total capital. V_{it}^{model} is the model-implied counterpart and $\bar{V}_t$ is the cross-sectional mean.

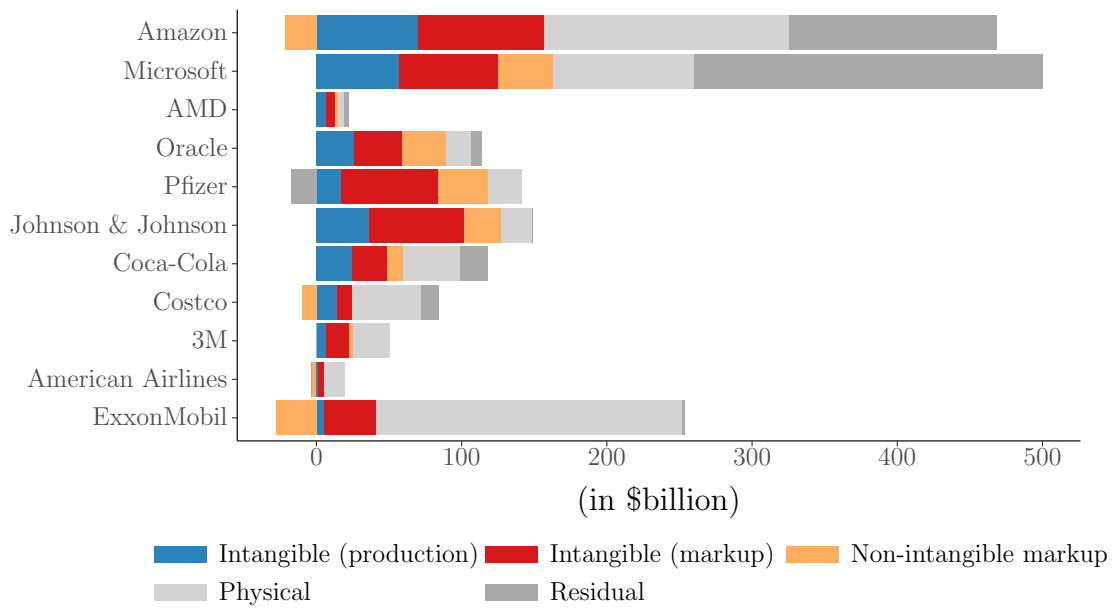


Figure 5: Decomposition for individual firms

This figure plots the value decomposition for a set of example firms based on Equation (18). Each bar represents a firm's average value across sample years.

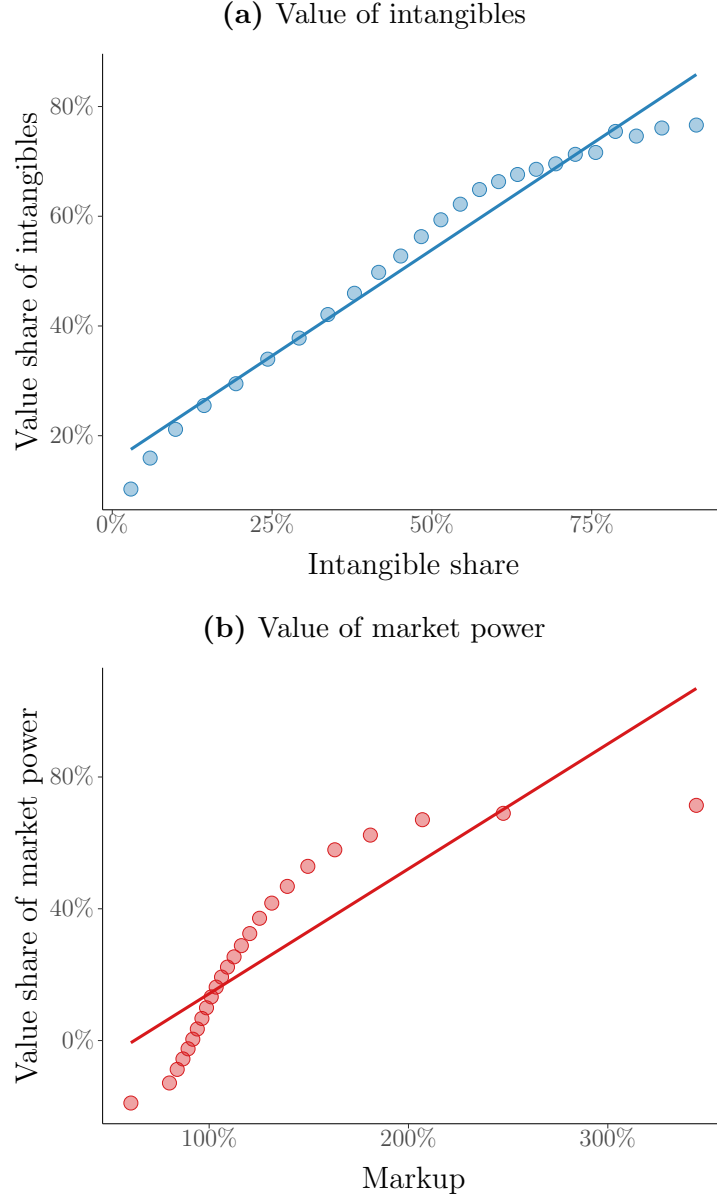


Figure 6: Value shares against intangible share and markup

This figure plots binned scatter plots of model-implied value shares against firm characteristics. Panel (a) plots the share of firm value attributable to intangibles ($V_{it}^{\text{prod-int}} + V_{it}^{\text{mkp-int}}$, scaled by V_{it}^{model}) against the firm's intangible share. Panel (b) plots the share of firm value attributable to market power ($V_{it}^{\text{mkp-int}} + V_{it}^{\text{mkp-non}}$, scaled by V_{it}^{model}) against the firm's markup. Observations are grouped into equal-sized bins based on the x -axis variable, and bin means are plotted. The solid lines show the linear fit.

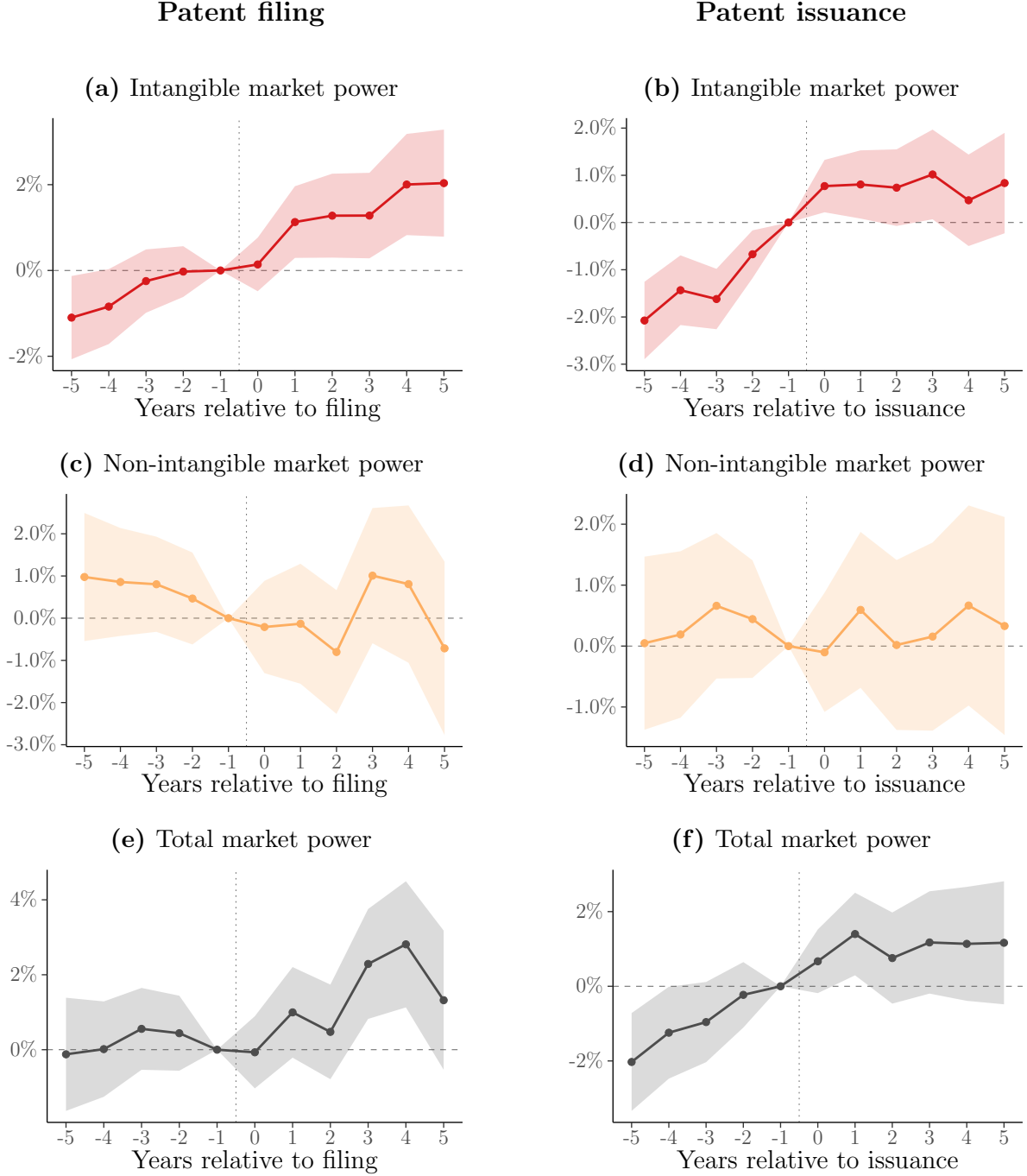


Figure 7: Change in the share of market power around patent filing and issuance

This figure plots dynamic treatment effects of important patent filings (left column) and issuances (right column) on the share of firm value attributable to market power. The outcomes are the intangible-driven market-power value share in Panels (a) and (b), the non-intangible market-power value share in Panels (c) and (d), and the total market-power value share in Panels (e) and (f). Each share is defined as the corresponding component divided by V^{model} and winsorized within each year at the 2.5th and 97.5th percentiles. The event-study design, patent-importance definition, and estimation follow Figure 2. Shaded areas are 95% confidence intervals, with standard errors clustered by firm.

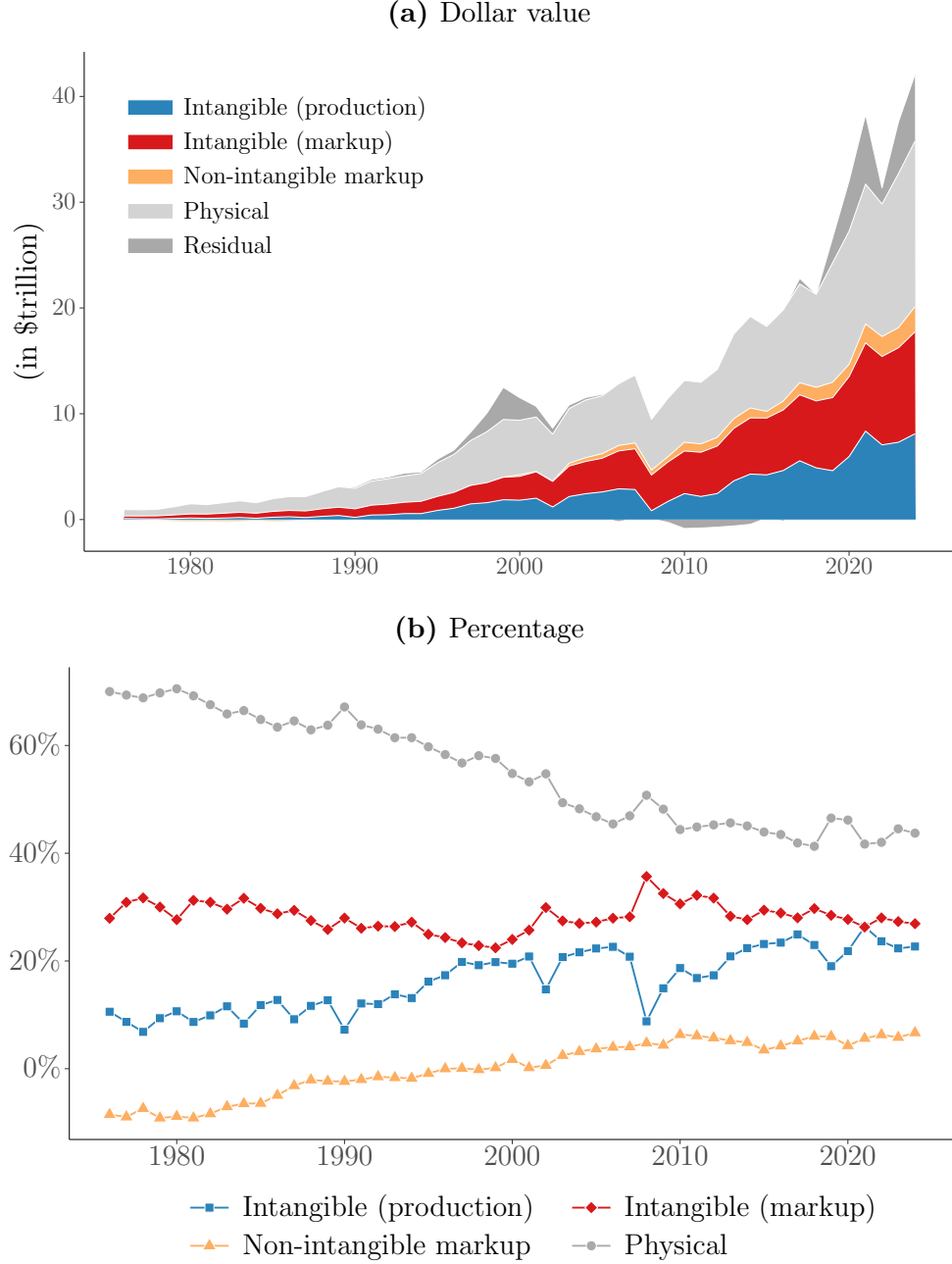


Figure 8: Level decomposition over time (1976–2024)

This figure plots the aggregate value decomposition based on Equation (18). Panel (a) sums each value component across all firms in the cross section for each year and reports the totals in trillions of dollars as a stacked area chart. Panel (b) shows the percentage decomposition, where each component's share is computed as its aggregate dollar value divided by total model-implied firm value.

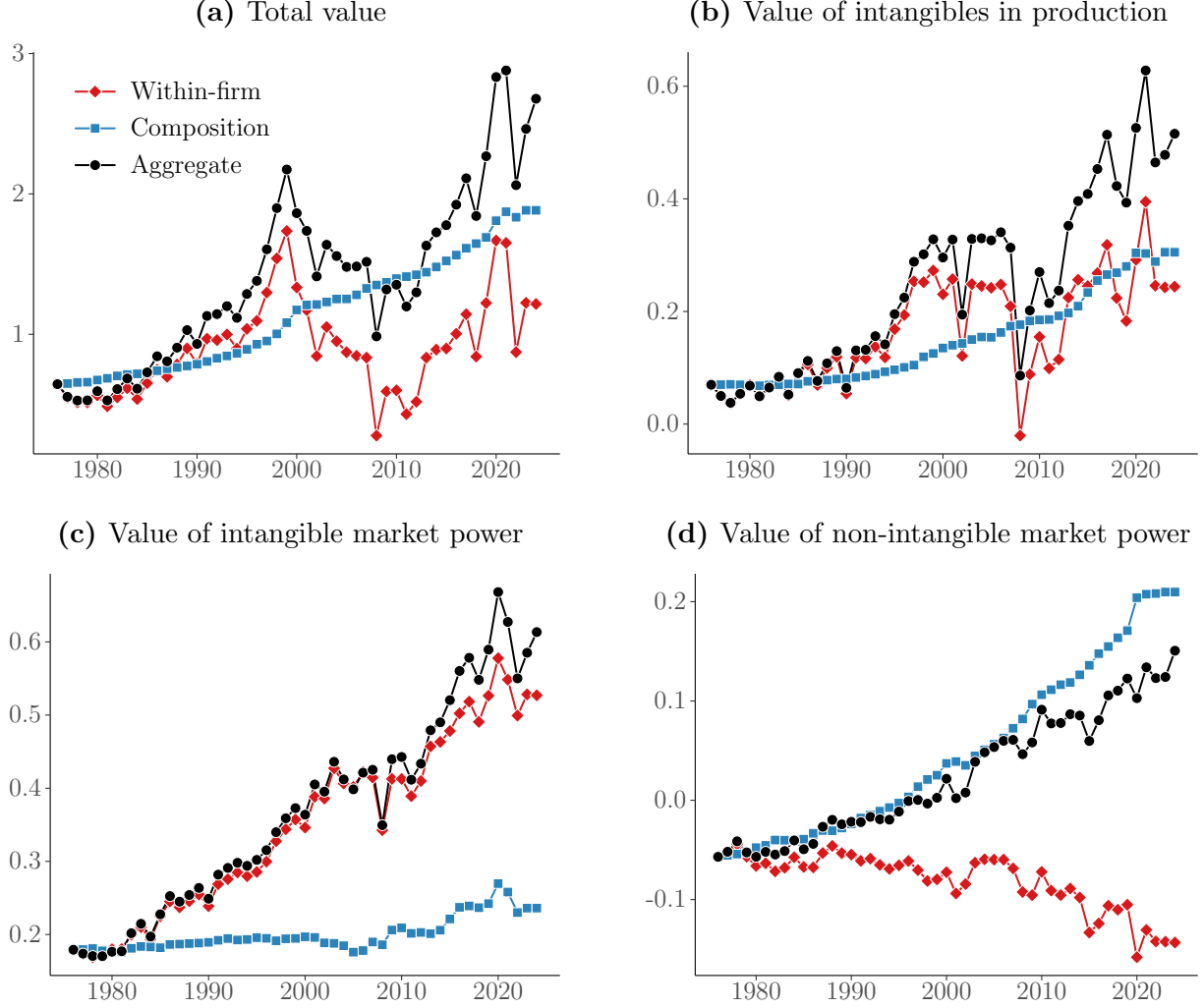


Figure 9: Compositional vs. within-firm change

This figure plots the decomposition of changes in aggregate value into within-firm and compositional components. The decomposition is defined in Equation (26). Panel (a) shows the decomposition of the total value-to-sales ratio. Panel (b) shows the decomposition of the value of intangibles in production scaled by sales. Panels (c) and (d) show the corresponding decompositions of intangible and non-intangible market-power value scaled by sales.

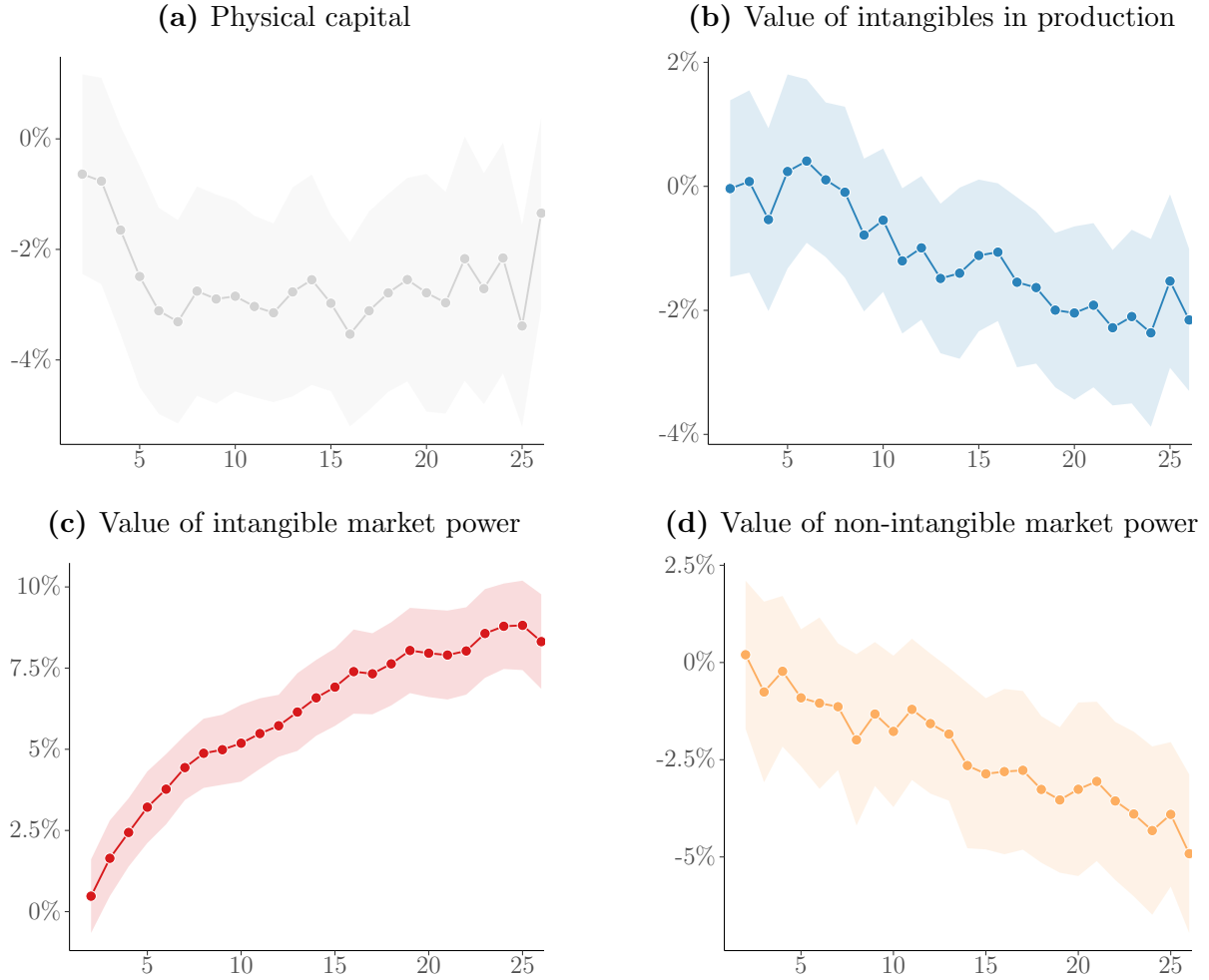


Figure 10: Life cycle changes

This figure plots the evolution of each value component's share over a firm's life cycle. For each component, I regress the ratio of the component to total model-implied value on firm age dummies with year and cohort fixed effects, restricting the sample to firms with at least 15 years of data. The solid line shows the estimated coefficients and the shaded area represents the 95% confidence interval.

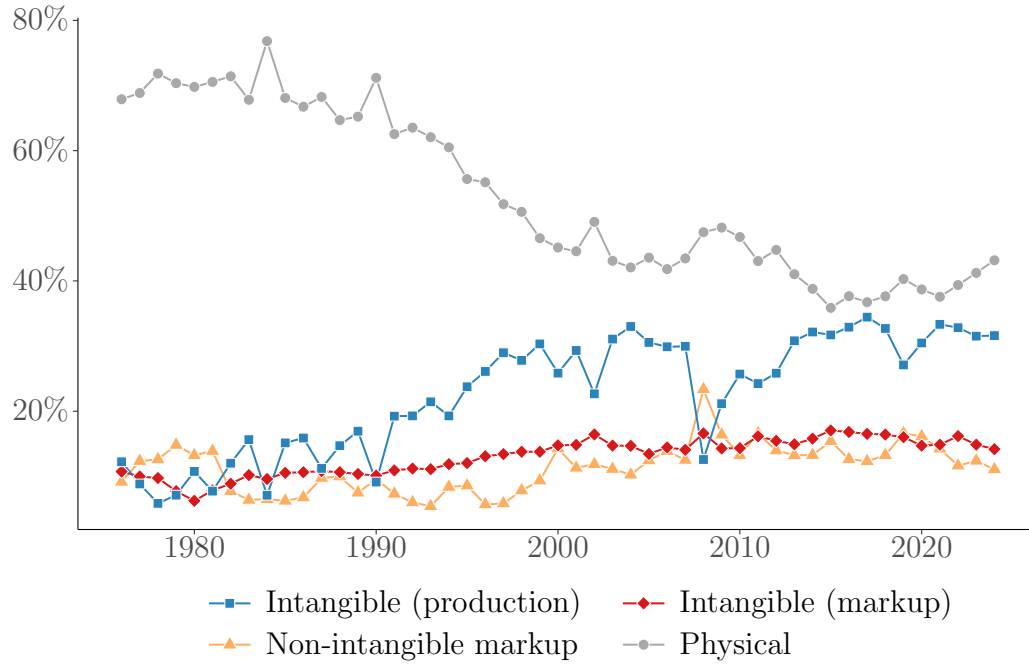


Figure 11: Covariance decomposition over time (1976–2024)

This figure plots the aggregate covariance decomposition for each year based on Equation (29).

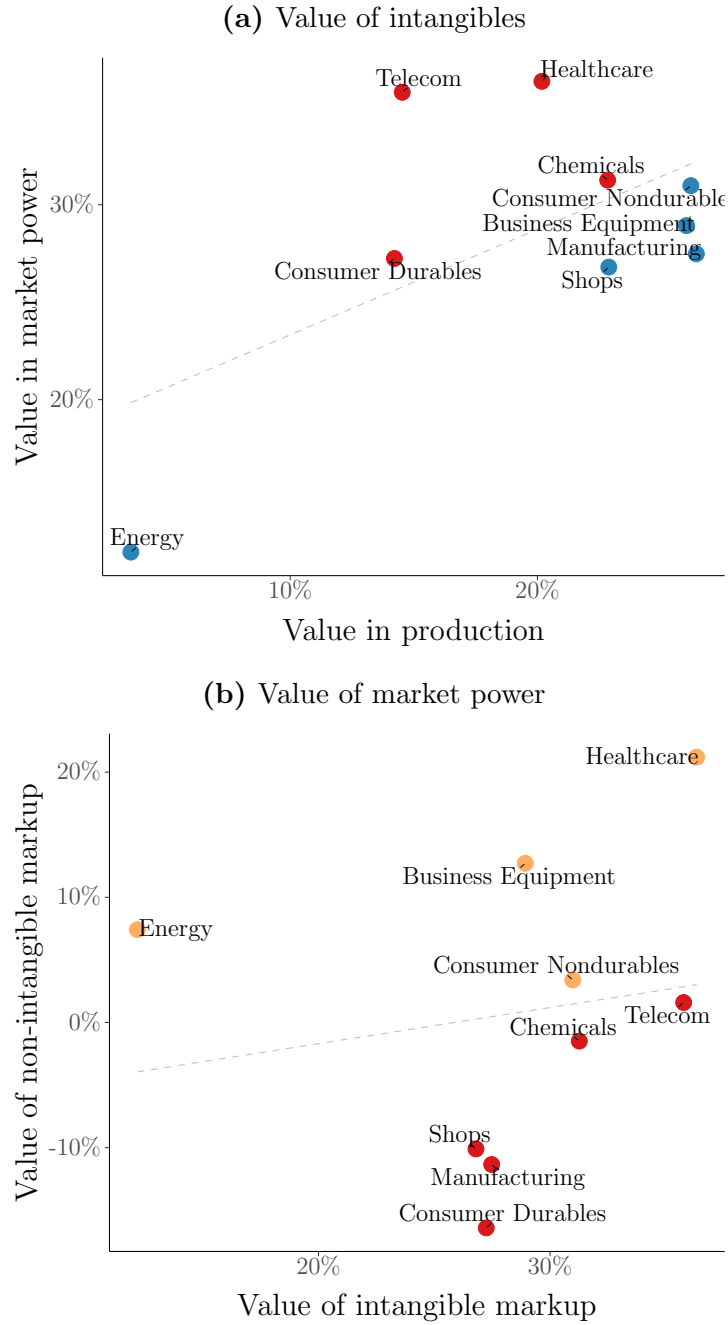


Figure 12: Sector-level relationships between value components

This figure plots sector-level scatter plots using the Fama-French 12 industry classification (excluding Financials, Utilities, and Other). Panel (a) plots the share of firm value attributable to intangible market power against the share from intangibles in production. Panel (b) plots the share from non-intangible market power against the share from intangible market power. Each point represents a sector average, and the dashed line shows the linear fit.

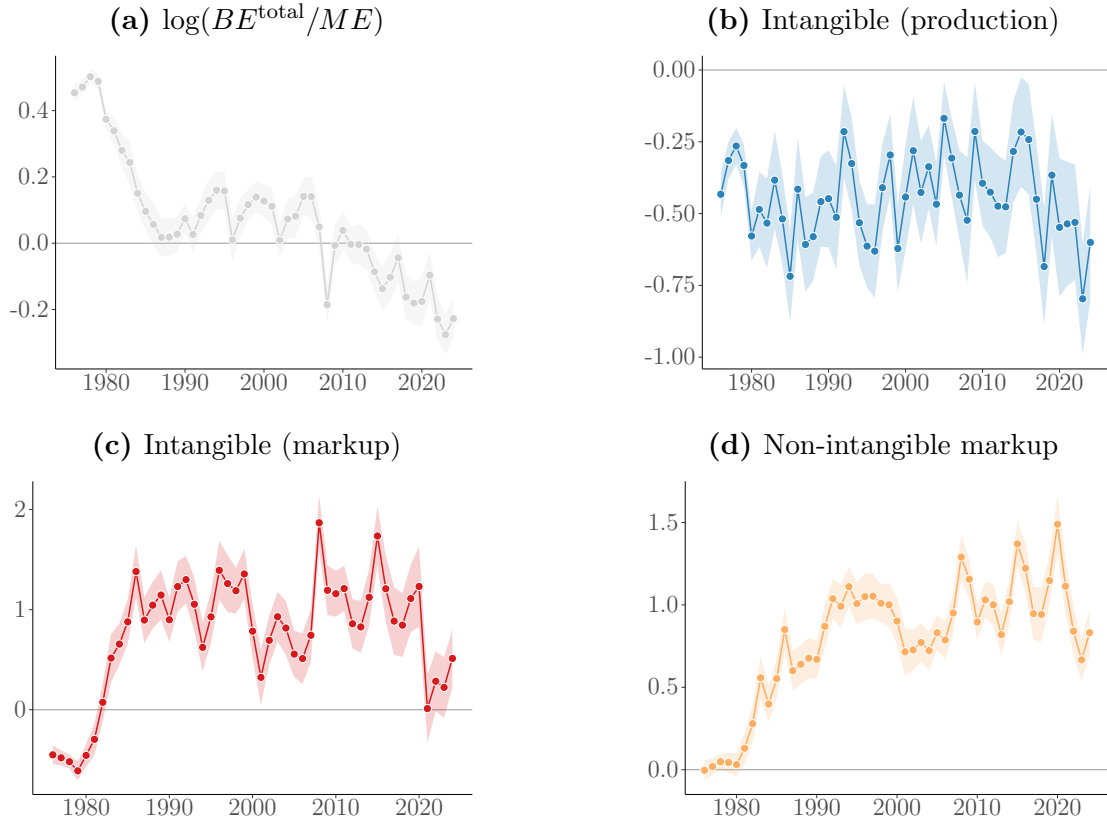


Figure 13: Annual coefficients in the fitted value signal regression

This figure plots the year-by-year coefficients of the cross-sectional regression in Equation (35), estimated separately on each year's cross-section. Panel (a) plots the coefficient on the log book-to-market ratio, and Panels (b), (c) and (d) the coefficients on the model-implied value shares of intangible production, intangible markup, and non-intangible markup. Shaded areas are 95% confidence intervals.

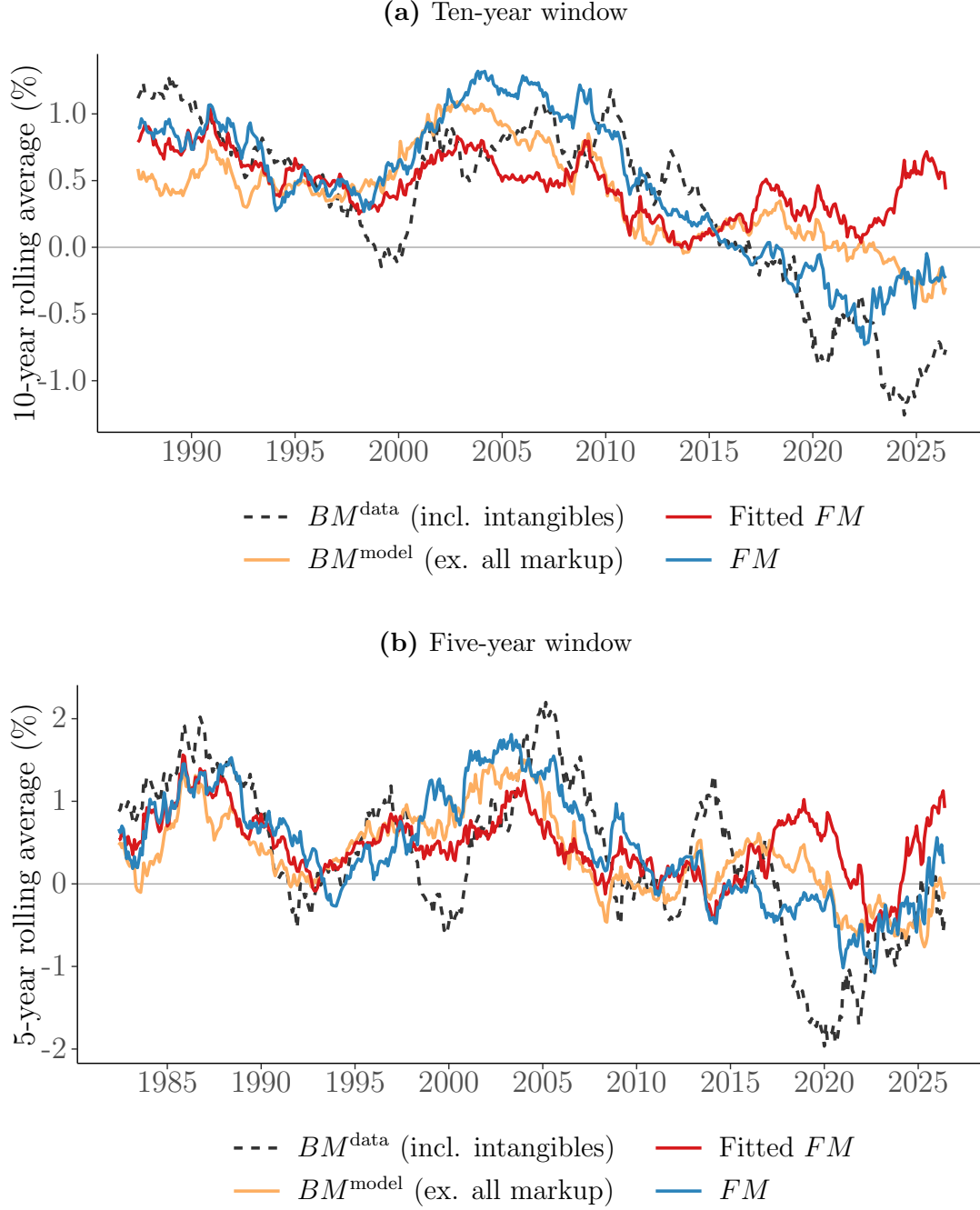


Figure 14: Trailing average of the value premium

This figure plots trailing averages of the monthly value premium, value minus growth in % per month, for four sorting variables: the intangible-adjusted book-to-market ratio, the model-implied ratio that removes the value of all market power, fitted FM , and FM . Portfolios are yearly NYSE-breakpoint deciles, value weighted by lagged market equity, over July 1977 to June 2026. Panel (a) uses a ten-year window and Panel (b) a five-year window. Windows are right-aligned, so the value plotted at June 2026 in Panel (a) is the average over July 2016 to June 2026.

Appendix: Derivation of the Operating Profits

A.1 Consumer Demand

Assume each firm operates within its own distinct market segment, producing a single, differentiated variety. The demand function faced by firm i is $Y(P_{it})$, where P_{it} is the price of product i . I parameterize $Y(\cdot)$ as

$$Y(P_{it}) = \left(\frac{P_{it}}{P_t} \right)^{-\frac{\mu_{it}}{\mu_{it}-1}} \frac{\bar{Y}}{P_t}, \quad \mu_{it} > 1, \quad (\text{A.1})$$

where μ_{it} represents the firm's pricing power for product i , $\bar{Y}$ is total market size for this specific product, and P_t is an exogenous price index that is absorbed by the productivity shocks (see the next section). Intuitively, when μ_{it} is high, consumers become less sensitive to prices and firms can set higher prices to extract rents. In the next subsection, I show that firms optimally choose markup to be μ_{it} in their profit maximization.

The total expenditure $\bar{Y}$ is held constant and serves as the numeraire in this market. This demand function implicitly assumes that all firms face the same demand curve, and demand variation comes from movement along the curve and variation of demand elasticity (through changing market power), while the total expenditure $\bar{Y}$ is constant across firms. This assumption is not too restrictive because I scale the firm size in the data to match the distribution in the model simulation and I match the valuation ratio to extract productivity states. However, the remaining unobserved demand-scale heterogeneity may still be absorbed by the recovered productivity states.

This demand function is consistent with a rational consumer preference system as shown in Matsuyama and Ushchev (2017) and Baqaee, Farhi and Sangani (2024), and it accommodates heterogeneous markups across firms. In this partial equilibrium framework, market power is captured through the shape of the demand function, which governs consumers' price sensitivity. This modeling choice does not imply that consumer preferences are the

sole primitive source of market power. In a general equilibrium setting, markups are equilibrium outcomes jointly determined by demand and supply forces: a firm's ability to charge high prices reflects not only consumers' willingness to pay but also supply-side factors such as product differentiation, proprietary technology, or competitive structure. The demand function in my model should therefore be interpreted as a reduced-form representation of these combined forces.

A.1.1 Special Cases of the Demand Function

The markup is less than or equal to 1. When $\mu_{it} \leq 1$, I assume the firm operates under perfect competition and takes the price P_t as given. This assumption corresponds to the limiting case of the demand function (A.1) as μ_{it} approaches 1 from above.

P_{it} is below the price index P_t . If the firm has the ability to produce with low marginal cost and undercut the price index, I assume the firm captures the entire market and earns total revenue $\bar{Y}$. This case corresponds to a sector monopoly as in De Ridder (2024) and Aghion et al. (2025), where the most efficient producer dominates the entire sector and earns constant revenue. The price this firm can set is constrained by the second-best competitor (potential entrant), and μ_{it} corresponds to the productivity gap between the incumbent and its closest rival. It is consistent with the relation between μ_{it} and intangible capital, as investments in intangibles enhance productivity and widen this competitive advantage.

A.2 Profit Maximization

At each time t , given the demand equation (A.1), firm i solves the following problem:

$$\begin{aligned} \Pi(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) &= \max_{P_{it}, L_{it}} P_{it} Y_{it} - P_{Lit} L_{it} \\ \text{subject to } & \mathbf{K}(z_{it}^K, z_{it}^N, K_{it}, N_{it})^\eta L_{it}^{1-\eta} = Y_{it}, \end{aligned} \tag{A.2}$$

where P_{Lit} is the price of the costlessly adjustable input and firms take it as given. Let λ_{it} be the Lagrangian multiplier for the production constraint. λ_{it} represents the marginal cost of production (given the capital inputs).

Taking the first-order condition with respect to P_{it} yields the optimal markup, $P_{it}/\lambda_{it} = \mu_{it}$. I then substitute P_{it} out using the constraint and rewrite the problem as

$$\max_{L_{it}} [\mathbf{K}_{it}^\eta L_{it}^{1-\eta}]^{\frac{1}{\mu_{it}}} \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}}} P_t^{\frac{1}{\mu_{it}}} - P_{Lit} L_{it}. \quad (\text{A.3})$$

The first-order condition with respect to L_{it} is $P_{Lit} = \frac{1-\eta}{\mu_{it}} \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}}} P_t^{\frac{1}{\mu_{it}}} Y_{it}^{\frac{1}{\mu_{it}}} / L_{it}$. Solving for L_{it} yields

$$L_{it} = \left(\frac{(1-\eta) \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}}} P_t^{\frac{1}{\mu_{it}}}}{\mu_{it} P_{Lit}} \right)^{\frac{\mu_{it}}{\mu_{it}-1+\eta}} \mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}. \quad (\text{A.4})$$

Substituting the first-order condition back into Π gives

$$\Pi_{it} = \left(1 - \frac{1-\eta}{\mu_{it}} \right) P_{it} Y_{it}. \quad (\text{A.5})$$

Plugging the optimal L_{it} into the profit function,

$$\begin{aligned} \Pi_{it} &= \left[1 - \frac{1-\eta}{\mu_{it}} \right] \left(\frac{1-\eta}{\mu_{it}} \right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} P_t^{\frac{1}{\mu_{it}-1+\eta}} \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}-1+\eta}} P_{Lit}^{-\frac{1-\eta}{\mu_{it}-1+\eta}} \mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}} \\ &= \left[1 - \frac{1-\eta}{\mu_{it}} \right] \left(\frac{1-\eta}{\mu_{it}} \right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} \bar{Y} \left(\frac{\mathbf{K}_{it}^\eta P_t}{P_{Lit}^{1-\eta} \bar{Y}^\eta} \right)^{\frac{1}{\mu_{it}-1+\eta}} \\ &= \left[1 - \frac{1-\eta}{\mu_{it}} \right] \left(\frac{1-\eta}{\mu_{it}} \right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} \bar{Y} \cdot \tilde{\mathbf{K}}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}. \end{aligned} \quad (\text{A.6})$$

The last equality is due to the fact that the term $\frac{P_t}{P_{Lit}^{1-\eta} \bar{Y}^\eta}$ can be absorbed by shocks z_{it}^K and z_{it}^N within the capital aggregator. In the main text, I drop the tilde above $\mathbf{K}$ to be more concise.

The demand function (A.1) is well-defined for $\mu_{it} > 1$. When $\mu_{it} = 1$ with price equal to

the marginal cost, the firm operates under perfect competition and is a price taker. In this case, the firm solves

$$\Pi(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) = \max_{L_{it}} P_t \mathbf{K}_{it}^\eta L_{it}^{1-\eta} - P_{Lit} L_{it}. \quad (\text{A.7})$$

The first-order condition is given by $P_{Lit} = (1 - \eta) P_t \mathbf{K}_{it}^\eta L_{it}^{-\eta}$, which implies $L_{it} = \left(\frac{(1-\eta)P_t}{P_{Lit}} \right)^{\frac{1}{\eta}} \mathbf{K}_{it}$. Therefore, the profit function is

$$\begin{aligned} \Pi_{it} &= P_t \mathbf{K}_{it}^\eta L_{it}^{1-\eta} - (1 - \eta) P_t \mathbf{K}_{it}^\eta L_{it}^{1-\eta} \\ &= \eta P_t \mathbf{K}_{it}^\eta L_{it}^{1-\eta} \\ &= \eta (1 - \eta)^{\frac{1-\eta}{\eta}} P_t^{\frac{1}{\eta}} P_{Lit}^{-\frac{1-\eta}{\eta}} \mathbf{K}_{it}, \end{aligned} \quad (\text{A.8})$$

which is equivalent to Equation (A.6) when setting $\mu_{it} = 1$. Therefore, the profit function (A.6) nests the perfect competition case.

When the firm produces at very low cost, it captures the entire product market niche and the profit is then given by (following Equation (A.5) and the fact that total sales equal the market size)

$$\Pi_{it} = \bar{Y} \cdot \left(1 - \frac{1 - \eta}{\mu_{it}} \right). \quad (\text{A.9})$$

Internet Appendix

“What Are Intangibles Worth? Production versus Market Power”

By Yicheng Liu

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A Data, Measurement, and Methodology Details

A.1 Measure of Markups

Estimation of firm-level markups typically relies on the production function approach in the IO literature (e.g., Hall, 1988; De Loecker, Eeckhout and Unger, 2020). This approach exploits the cost-minimization condition that ties marginal cost to input demand. Markups are then estimated as the wedge between the output elasticity of a variable input and that input’s observed revenue share. The measure is given by

$$\mu_{it} = \eta_{it} \frac{\text{SALE}_{it}}{\text{VarCost}_{it}}, \quad (\text{A.1})$$

where VarCost_{it} is the cost of a variable input that can be freely adjusted, and η_{it} is the output elasticity of that input.

While this measure provides markup estimates for nearly all public firms in Compustat, it is criticized for relying on strong assumptions: the production function can be identified from revenue-based data when quantity is not observed, and COGS from Compustat fully represents the variable inputs.¹³ However, De Ridder, Grassi and Morzenti (2026) argue that even though the level of markup can be biased under the production approach, the dispersion among firms is accurately captured by such estimates, which is the focus of this paper. To further ensure the validity of this measure, I include 6 estimates based on different specifications.

Model-free markup proxies. As shown in De Ridder, Grassi and Morzenti (2026), firm-level variation in markups is primarily driven by the variation in the revenue-to-cost ratio $\text{SALE}_{it}/\text{VarCost}_{it}$. I first document the correlation between intangible capital share and this ratio, which is free of production function estimation. I consider two versions of VarCost_{it} .

¹³See Traina (2018), Bond et al. (2021), Doraszelski and Jaumandreu (2022), Raval (2023), and Demirer (2022).

The first uses COGS following De Loecker, Eeckhout and Unger (2020). The second includes both COGS and some of the SG&A expenses. Traina (2018) suggests that at least some of the selling and management expenses are also variable costs. I follow Ayyagari, Demirgüç-Kunt and Maksimovic (2024) to measure variable costs as:

$$\text{COGS} + \text{XSGA}^* = \text{COGS} + (\text{XSGA} - \text{XRD} - \text{RDIP} - 0.3 \times \text{XSGA}), \quad (\text{A.2})$$

where XSGA is the Compustat item selling, general, and administrative expenses. XRD and RDIP measure R&D and in-process R&D expenses in Compustat. In other words, this measure of variable costs recognizes R&D and some SG&A expenses as capital investment, while the rest is variable.

Markup based on different production functions. I further estimate markups under four production function specifications, varying along two dimensions: production technology (translog vs. Cobb-Douglas) and whether intangible capital is included as a production input. The translog production function approximates arbitrary production technologies via a second-order Taylor expansion in logarithms. In general, a translog function with n inputs $X_1, X_2, \dots, X_n$ can be written as:

$$\log Y = \alpha_0 + \sum_{i=1}^n \alpha_i \log X_i + \sum_{i=1}^n \sum_{j=1}^n \beta_{ij} \log X_i \log X_j. \quad (\text{A.3})$$

Under the translog specification, the output elasticity of the variable input, $\eta_i = \partial \log Y / \partial \log X_i$, is firm-specific and depends on the levels of all production inputs. Cobb-Douglas is a nested special case with $\beta_{ij} = 0$ for all i, j , yielding a constant output elasticity. For each production technology, I consider two input configurations: one with physical capital and the variable input only, and one that additionally includes intangible capital. In all specifications, I use cost of goods sold (COGS) as the variable input expenditure.

A.2 Estimation Details

The SMM estimation minimizes the distance between data moments and simulated moments. The estimation is implemented in the following steps. For an arbitrary set of parameters Θ , I solve the model for the value and policy functions. Then I simulate K panels of N firms $\times T + 150$ years and drop the first 150 years as a burn-in period to remove the impact of initial states. In the simulation, if a firm's value drops to 0, I assume the firm exits and is immediately replaced by a new firm with the same capital and redrawn shocks.

Next, I compute the model-implied moments in the simulation. Let x_{it} be the actual data vector, $i \in \{1, \dots, N\}$, $t \in \{1, \dots, T\}$, and let $y_{itk}(\Theta)$ be the simulated vector from simulation k , $i \in \{1, \dots, N\}$, $t \in \{1, \dots, T\}$, and $k \in \{1, \dots, K\}$. The simulated data vector, $y_{itk}(\Theta)$, depends on a vector of structural parameters, Θ . Define the moment conditions as

$$\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \left[h(x_{it}) - \frac{1}{K} \sum_{k=1}^K h(y_{itk}(\Theta)) \right] \equiv \Psi^A - \Psi^S(\Theta), \quad (\text{A.4})$$

where $h(y_{itk}(\Theta))$ is a vector of simulated moments and $h(x_{it})$ is the vector of moments from the actual data. $\Psi^A = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T h(x_{it})$, $\Psi^S(\Theta) = \frac{1}{NTK} \sum_{i=1}^N \sum_{t=1}^T \sum_{k=1}^K h(y_{itk}(\Theta))$.

The simulated moments estimator is defined as the solution to the minimization of

$$\hat{\Theta} = \arg \min_{\Theta} [\Psi^A - \Psi^S(\Theta)]' \hat{W} [\Psi^A - \Psi^S(\Theta)], \quad (\text{A.5})$$

where $\hat{W}$ is a positive definite matrix that converges in probability to a deterministic positive definite matrix W . It is constructed by calculating the inverse of the variance-covariance matrix of the data moments. I construct the variance-covariance matrix using the influence function method following Erickson and Whited (2002). I demean each variable at the industry-year level and then calculate the influence functions for each moment using the demeaned data. I then covary these influence functions by summing in both the cross-sectional and time series dimensions.

I use the simulated annealing algorithm to solve the minimization problem. The algorithm starts with an initial guess of parameters and iteratively updates the parameters by adding a random perturbation. The new parameters are accepted with a probability that depends on the change in the objective function and a temperature parameter that gradually decreases over iterations. This allows the algorithm to explore the parameter space and avoid local minima. I set the initial temperature to be sufficiently high to allow for exploration and also start with different initial guesses to ensure the robustness of the estimation results.

The simulated moments estimator is asymptotically normal for fixed K . Denote $g(\Theta) \equiv \Psi^A - \Psi^S(\Theta)$, let Θ_0 be the true parameter vector, and define $G \equiv \partial g(\Theta_0)/\partial \Theta'$. The asymptotic distribution of $\hat{\Theta}$ is given by:

$$\sqrt{n}(\hat{\Theta} - \Theta_0) \xrightarrow{d} \mathcal{N}\left(0, \text{avar}(\hat{\Theta})\right), \quad (\text{A.6})$$

where

$$\text{avar}(\hat{\Theta}) = \left(1 + \frac{1}{K}\right) (G'WG)^{-1} G'W\Omega WG(G'WG)^{-1}, \quad (\text{A.7})$$

where Ω is the asymptotic covariance matrix of the moment vector. I estimate this covariance matrix using the influence function of the moment vector, clustered at the firm level, following Erickson and Whited (2002).

A.3 Shapley Decomposition

As described in Section 4.3, separating the production and markup channels of intangible capital requires treating N^{prod} and N^{mkp} as distinct variables. Since the original value function V has only a single state variable N , I solve an auxiliary value function $\tilde{V}$ from a model with the same parameters but two separate intangible capital stocks: N^{prod} enters only the production function and N^{mkp} enters only the markup function. One concern is that in the auxiliary problem, firms need to invest in two copies of intangible capital to achieve the same level of production and market power as in the original problem. To ensure that the two

problems have a comparable cost structure, I set the adjustment cost as a function of total investment rates in two intangibles, which is given by $\frac{b^N}{2} \left(\frac{I_{it}^{N,\text{prod}} + I_{it}^{N,\text{mkp}}}{N_{it}^{\text{prod}} + N_{it}^{\text{mkp}}} \right)^2 (N_{it}^{\text{prod}} + N_{it}^{\text{mkp}})$, where $I_{it}^{N,\text{prod}}$ and $I_{it}^{N,\text{mkp}}$ are the investments in the two intangible capital stocks.

I then define a composite evaluation rule $\hat{V}$ that governs how coalitions are evaluated throughout the Shapley decomposition:

$$\hat{V}(\cdot) = \begin{cases} V(\cdot) & \text{if } N^{\text{prod}} = N^{\text{mkp}} \\ \tilde{V}(\cdot) & \text{if } N^{\text{prod}} \neq N^{\text{mkp}}. \end{cases} \quad (\text{A.8})$$

That is, when N^{prod} and N^{mkp} are at the same value—both at benchmark or both at their firm-level value—the coalition is evaluated using the original V . The auxiliary $\tilde{V}$ is used only when exactly one of the two has been switched on. This ensures that the decomposition recovers the original V_{it} at the grand coalition while allowing the production and markup channels to be separated.

I then run a Shapley decomposition over six variables $\mathcal{N} = \{z^K, z^N, \zeta^\mu, K, N^{\text{prod}}, N^{\text{mkp}}\}$. For a given variable $j \in \mathcal{N}$ and a subset $S \subseteq \mathcal{N} \setminus \{j\}$, define $\hat{V}(S \cup \{j\})$ as the composite value function evaluated with variable j and every variable in S set to their firm-level values, and all remaining variables set to their benchmark values. Similarly, $\hat{V}(S)$ sets every variable in S to its firm-level value and all other variables (including j) to their benchmark values. The Shapley value of variable j is:

$$V_{it}^j = \sum_{S \subseteq \mathcal{N} \setminus \{j\}} \frac{|S|! (5 - |S|)!}{6!} \left[\hat{V}(S \cup \{j\}) - \hat{V}(S) \right], \quad (\text{A.9})$$

where $|S|$ denotes the cardinality of S . The sum ranges over all $2^5 = 32$ subsets of the five variables other than j , weighting each marginal contribution by the Shapley coefficients. The empty coalition corresponds to all variables at their benchmark values, yielding $\hat{V}(\emptyset) = \bar{V}$, and the grand coalition corresponds to all variables at their firm-level values, yielding $\hat{V}(\mathcal{N}) = V_{it}$, since $N^{\text{prod}} = N^{\text{mkp}} = N_{it}$ at the grand coalition.

Intuitively, the Shapley decomposition calculates the average change in firm value when a variable moves from its benchmark to its firm-level value, across all possible orderings in which the six variables are added one by one. I set the benchmark productivity and exogenous markup shocks to zero and the benchmark capital stocks to their minimum grid values. The decomposition term $V_{it}^{\zeta^\mu}$ captures the marginal contribution of the exogenous markup shock relative to its zero-shock benchmark, averaged across all possible orderings of the other variables. The term $V_{it}^{N,\text{prod}}$ measures the value of adding intangible capital to the production process only, without its effect on markup, since N^{mkp} is treated as a separate variable. Similarly, $V_{it}^{N,\text{mkp}}$ captures the value of intangible capital's contribution to market power, holding the production channel fixed.

A.4 Estimation of state variables

This appendix details the iterative procedure for estimating the productivity terms z_{it}^K and z_{it}^N in Section 4.3. Each is the sum of an aggregate and a firm-specific component, $z_{it}^j = x_t^j + \zeta_{it}^j$ for $j \in \{K, N\}$, governed by different distributions (Section 3.3). Inverting the valuation moments in Equation (20) firm-by-firm recovers only the sum z_{it}^j . The aggregate component is the cross-sectional average of the firm-level productivity in each year, $x_t^j = |\mathcal{I}_t|^{-1} \sum_{i \in \mathcal{I}_t} z_{it}^j$, and each firm's idiosyncratic component is the residual $\zeta_{it}^j = z_{it}^j - x_t^j$. I use bounds based on the model-implied unconditional volatilities to rule out extreme values of both components. Because each firm's admissible range is centered on the year's aggregate, which is itself the average of the firm-level estimates, I impose these bounds iteratively.

For each capital type j , the aggregate and idiosyncratic components follow AR(1) processes with common persistence ρ^j and innovation standard deviations σ_x^j and σ_ζ^j , so their unconditional standard deviations are $s_x^j = \sigma_x^j / \sqrt{1 - (\rho^j)^2}$ and $s_\zeta^j = \sigma_\zeta^j / \sqrt{1 - (\rho^j)^2}$. I bound the aggregate component by $|x_t^j| \leq 3 s_x^j$ and the idiosyncratic component by $|\zeta_{it}^j| \leq 3 s_\zeta^j$, and keep every estimate within the model's productivity grid $[\underline{z}^j, \bar{z}^j]$.

Let $\mathcal{I}_t$ be the set of firms observed in year t . Starting from $x_t^{j,(0)} = 0$ for every year, I

iterate two steps until convergence. At iteration m :

Step 1 (Estimation). Taking $x_t^{j,(m-1)}$ as given, for each firm-year I solve Equation (20), restricting each productivity term to the intersection of its idiosyncratic and grid bounds,

$$z_{it}^j \in \left[x_t^{j,(m-1)} - 3 s_\zeta^j, x_t^{j,(m-1)} + 3 s_\zeta^j \right] \cap [\underline{z}^j, \bar{z}^j], \quad j \in \{K, N\}, \quad (\text{A.10})$$

so a firm's estimate may deviate from the year's aggregate by at most three unconditional standard deviations. I obtain a starting value by grid search over the admissible rectangle and refine it with a local (Nelder–Mead) search.

Step 2 (Aggregation). I update each year's aggregate to the cross-sectional average of the firm-level estimates, bounded by $3 s_x^j$:

$$x_t^{j,(m)} = \frac{1}{|\mathcal{I}_t|} \sum_{i \in \mathcal{I}_t} \hat{z}_{it}^j, \quad \text{bounded to } |x_t^{j,(m)}| \leq 3 s_x^j, \quad j \in \{K, N\}. \quad (\text{A.11})$$

I stop when $\max_t (|x_t^{K,(m)} - x_t^{K,(m-1)}| + |x_t^{N,(m)} - x_t^{N,(m-1)}|) < 5 \times 10^{-3}$, or when successive changes stagnate. At the fixed point I set $\hat{x}_t^j = x_t^{j,*}$ and recover each firm's idiosyncratic productivity as $\hat{\zeta}_{it}^j = \hat{z}_{it}^j - \hat{x}_t^j$. This procedure keeps both components within ranges implied by their model-implied unconditional volatilities.

B Additional Results

This section organizes the supplemental empirical results by topic. Table [IA.1](#) collects the variable definitions used across these analyses.

B.1 Additional Evidence on Intangibles and Markups

Figures [IA.1](#) and [IA.2](#) test whether the positive relation between intangible intensity and markups is sensitive to the markup measure. They repeat the annual correlations and binned-scatter plots for two model-free sales-to-cost ratios and four production-function estimates that vary the functional form and inclusion of intangible capital. Table [IA.2](#) reports corresponding regressions with firm controls and industry and year fixed effects. The positive association remains across all six measures.

Figure [IA.3](#) uses a stacked difference-in-differences design that treats every important patent as an event. Figures [IA.4–IA.7](#) vary the markup measure, patent definition, and event date, covering filings and issuances for highly cited and high-value patents under all six markup measures. Because the stacked estimates display stronger pre-trends, I treat them as robustness evidence rather than the preferred design.

B.2 Value Shares of Market Power and Industry Dynamics

Table [IA.3](#) relates the intangible and non-intangible market-power value shares to standard measures of product-market dynamics. Panel A aggregates the value shares to NAICS-3 industry-year cells and considers concentration and establishment net entry, while Panel B uses firm-level TNIC concentration, product similarity, and fluidity within NAICS-3 industry-year cells. The components have different associations with these measures, but the evidence is descriptive rather than causal.

B.3 Alternative Specification for Life Cycle Regressions

Figure [IA.8](#) replaces the age indicators in the baseline life-cycle regressions with natural cubic splines, while retaining the sample restriction to firms observed for at least 15 years and the year and cohort fixed effects. The smoother specification yields the same qualitative reallocation within intangible value: the production component declines with firm age, whereas intangible-driven market power rises.

B.4 Additional Evidence on the Value Premium Decline

Table [IA.4](#) replaces the continuous time-trend specification with early- and late-sample portfolio returns based on three alternative split dates: July 1996, January 2002, and January 2010. The magnitude and significance of the early-late comparison vary with the selected date, illustrating why the main analysis uses the full monthly series rather than a single sample split. Across the alternatives, the fitted fundamental-to-market signal continues to deliver the strongest value premium in the second subsample.

B.5 Alternative Definitions of Investment Rates and q

Table [IA.5](#) uses the exact Peters–Taylor definition of total q , rather than the paper’s cash-adjusted measure, and tests whether it retains explanatory power after the four model-implied value components are included. This specification checks that the component results do not depend on the paper’s particular construction of observed total q .

Tables [IA.6](#) and [IA.7](#) instead scale total, physical, and intangible investment—and the corresponding value measures—by total capital $K + N$. This common denominator makes coefficients comparable across investment margins but also introduces variation in capital composition. The horse races, component regressions, and Shapley decomposition remain qualitatively similar to the baseline results that scale each investment rate by its corresponding capital stock.

Table IA.1: Variable definitions

Variable	Definition
Intangible capital	Accumulated R&D and capitalized SG&A using the perpetual-inventory method, plus balance-sheet goodwill, from Peters and Taylor (2017)
Physical capital	PPEGT
Market equity	Price (CRSP item PRC) $\times$ shares outstanding (SHROUT)
Total firm value	Market equity + DLC + DLTT – CHE
Profitability	$(\text{REVT} - \text{COGS}) / (\text{PPEGT} + \text{intangible capital})$
Physical investment rate	CAPX / PPEGT
Intangible investment rate	$(\text{XRD} + \text{capitalized share of XSGA}) / \text{intangible capital}$
Intangible share	$\text{Intangible capital} / (\text{PPEGT} + \text{intangible capital})$

Table IA.2: Relation between intangible share and markups

This table reports OLS regressions of markups on the intangible capital share and firm-level controls. The dependent variable in each column is a different markup measure: column (1) uses SALE/COGS; column (2) uses SALE/(COGS + XSGA*); columns (3) and (4) use Cobb-Douglas production function estimates without and with intangible capital as a production input; columns (5) and (6) use translog production function estimates without and with intangible capital. Intangible share is intangible capital divided by the sum of physical and intangible capital. Leverage is total debt divided by total assets. ROA is return on assets. The sample is the model decomposition sample used throughout the paper. All specifications include two-digit industry and year fixed effects. Standard errors in parentheses are two-way clustered by firm and year.

	(1) $\frac{\text{SALE}}{\text{COGS}}$	(2) $\frac{\text{SALE}}{\text{COGS}+\text{XSGA}^*}$	(3) CD	(4) CD (intan.)	(5) TL	(6) TL (intan.)
Intan. share	1.1880*** (0.0730)	0.2435*** (0.0255)	1.0881*** (0.0614)	0.9750*** (0.0543)	1.0806*** (0.0615)	0.9903*** (0.0576)
Log age	-0.1227*** (0.0134)	-0.0329*** (0.0047)	-0.1107*** (0.0108)	-0.0941*** (0.0095)	-0.1104*** (0.0109)	-0.1085*** (0.0103)
ROA	1.0849*** (0.1101)	0.7200*** (0.0765)	0.7369*** (0.0923)	0.6737*** (0.0847)	0.7283*** (0.0896)	0.6583*** (0.0839)
Leverage	-0.1295** (0.0500)	0.0083 (0.0266)	-0.1364*** (0.0390)	-0.1070*** (0.0351)	-0.1346*** (0.0385)	-0.1449*** (0.0360)
Log assets	0.0047 (0.0063)	0.0208*** (0.0027)	0.0045 (0.0052)	0.0030 (0.0046)	-0.0083 (0.0051)	-0.0328*** (0.0048)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	122,670	115,089	109,846	109,844	109,846	109,844
Adj. R^2	0.2481	0.3311	0.2357	0.2347	0.2437	0.2156
Within R^2	0.1105	0.2041	0.1219	0.1251	0.1204	0.1226

Table IA.3: Market-power value shares and industry dynamics

This table reports panel regressions of the intangible and non-intangible market-power value shares on product-market competition measures. Each column is a separate univariate regression of a value share on one competition measure. Panel A is estimated at the industry level: the value shares are aggregated to NAICS-3 industry-year cells (value-weighted, the ratio of aggregate markup value to aggregate firm value in the industry) and regressed on industry-level competition measures with year fixed effects. The measures are the sales Herfindahl (HHI) and the four-firm concentration ratio (CR4), both computed at the NAICS-3 level, and the Census Business Dynamics Statistics establishment net entry rate (entry minus exit). Panel B is estimated at the firm level with NAICS-3 industry-by-year fixed effects. The measures are the Hoberg and Phillips (2016) TNIC Herfindahl (TNIC HHI), TNIC similarity, and product-market fluidity (the change in rivals' product vocabulary relative to the firm's own). All Panel B regressions control for firm size (log sales). Standard errors in parentheses are two-way clustered by industry and year in Panel A and by firm and year in Panel B.

Panel A: Industry-level product market measures (1976–2024)						
	Intangible market-power share			Non-intangible market-power share		
	HHI	CR4	Net entry	HHI	CR4	Net entry
Coefficient	0.0038 (0.0277)	-0.0118 (0.0331)	-0.0055*** (0.0016)	-0.0306 (0.0617)	-0.1024* (0.0552)	0.0069** (0.0028)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,013	4,013	3,339	4,013	4,013	3,339
R^2	0.0864	0.0869	0.1095	0.0293	0.0449	0.0296
Within R^2	0.0001	0.0006	0.0221	0.0015	0.0176	0.0117
Panel B: Firm-level product market measures (1988–2023)						
	Intangible market-power share			Non-intangible market-power share		
	TNIC HHI	Similarity	Fluidity	TNIC HHI	Similarity	Fluidity
Coefficient	0.0207*** (0.0061)	0.0004 (0.0003)	-0.0014** (0.0005)	-0.0443*** (0.0101)	-0.0011* (0.0006)	0.0011 (0.0009)
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm size control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	89,879	89,879	87,461	89,879	89,879	87,461
R^2	0.3122	0.3114	0.3127	0.2876	0.2868	0.2867
Within R^2	0.0179	0.0166	0.0178	0.0076	0.0065	0.0061

Table IA.4: Value premiums by sample split date

This table reports the value premium, value minus growth in % per month, estimated separately in the first and second subperiods of the sample for three candidate split dates. The sorts are identical to those in Panel A of Table 6: yearly NYSE-breakpoint decile sorts, value weighted by lagged market equity. The full return sample runs from July 1977 to June 2026, so for each split date the first subperiod runs from July 1977 to the month before the split and the second subperiod from the split month to June 2026; the two subperiods for each split date therefore partition the full sample exactly. Newey–West t -statistics using automatic lag selection are in parentheses beneath the estimates. For the BM ratios, the value portfolio is the low market-to-book decile; for fitted FM and FM , it is the high decile.

Split date	July 1996		January 2002		January 2010	
	First	Second	First	Second	First	Second
<i>Data ratios</i>						
BM^{data} (original)	0.49 (1.40)	0.01 (0.02)	0.43 (1.40)	-0.04 (-0.09)	0.51 (1.86)	-0.44 (-0.81)
BM^{data} (incl. intangibles)	0.74 (2.03)	0.06 (0.19)	0.71 (2.09)	-0.05 (-0.15)	0.68 (2.31)	-0.37 (-0.83)
<i>Model-implied ratios</i>						
BM^{model} (ex. exogenous markup)	0.41 (1.57)	-0.04 (-0.20)	0.45 (1.80)	-0.18 (-0.92)	0.34 (1.55)	-0.26 (-1.26)
BM^{model} (ex. all markup)	0.49 (2.18)	0.23 (1.11)	0.67 (3.07)	-0.01 (-0.04)	0.54 (2.87)	-0.10 (-0.38)
<i>Fundamental-to-market</i>						
Fitted FM	0.63 (2.64)	0.40 (1.96)	0.63 (3.31)	0.35 (1.50)	0.57 (3.29)	0.34 (1.11)
FM	0.61 (2.25)	0.34 (1.30)	0.83 (3.37)	0.06 (0.23)	0.77 (3.81)	-0.19 (-0.61)

Table IA.5: Investment and q : Peters-Taylor q and the value components

This table asks whether Peters-Taylor q still explains investment after controlling for the model's four value components. Columns (1)–(2) explain physical investment I^K/K and columns (3)–(4) intangible investment I^N/N ; both dependent variables are measured one year ahead, so the regression is of investment in $t + 1$ on q in t . Columns (1) and (3) use Peters-Taylor q on its own, and columns (2) and (4) add the four value components alongside it. The components are scaled by the capital stock of the margin being explained: by K in column (2) and by N in column (4). The sample is restricted to the firm-years for which Peters-Taylor q is available, and is therefore slightly smaller than in the other investment tables. All specifications include firm and year fixed effects, with standard errors two-way clustered by firm and year in parentheses.

	Physical, I^K/K		Intangible, I^N/N	
	(1)	(2)	(3)	(4)
Peters-Taylor q	0.0346*** (0.0032)	0.0260*** (0.0028)	0.0304*** (0.0023)	0.0269*** (0.0022)
V^{phy}		0.0124*** (0.0020)		0.0010** (0.0004)
$V^{\text{prod-int}}$		-0.0015 (0.0011)		0.0000 (0.0026)
$V^{\text{mkp-int}}$		0.0178*** (0.0027)		0.0350*** (0.0066)
$V^{\text{mkp-non}}$		-0.0035** (0.0013)		0.0011** (0.0005)
Observations	105,747	105,747	105,747	105,747
R^2	0.4686	0.4834	0.6238	0.6357
Within R^2	0.1167	0.1413	0.1242	0.1520

Table IA.6: Investment and q : horse race, scaled by total capital

This table repeats Table 8 with every investment rate scaled by total capital $K+N$ rather than by the capital stock of its own margin, which makes the three margins comparable on a common base. Each column is a separate regression of the investment rate on a single q , so the coefficients in a panel are directly comparable across columns. Panel A uses total investment $I/(K+N)$, Panel B uses physical investment $I^K/(K+N)$, and Panel C uses intangible investment $I^N/(K+N)$; all three are led one year, so the regression is of investment in $t+1$ on q in t . Model total q is the model-implied value over $K+N$; the last two columns strip the non-intangible market-power component $V^{\text{mkp-non}}$, and then both market-power components $V^{\text{mkp-int}}$ and $V^{\text{mkp-non}}$, out of that value. All specifications include firm and year fixed effects, with standard errors two-way clustered by firm and year in parentheses.

	Standard q	Total q	Model total q	Model total q (ex. non-intangible markup)	Model total q (ex. all markup)
	(1)	(2)	(3)	(4)	(5)
Panel A: Total investment, $I/(K+N)$					
q	0.0047*** (0.0004)	0.0308*** (0.0027)	0.0618*** (0.0031)	0.0637*** (0.0031)	0.0694*** (0.0033)
Observations	105,979	105,979	105,979	105,979	105,979
R^2	0.5665	0.6182	0.6154	0.6109	0.6105
Within R^2	0.0769	0.1869	0.1811	0.1715	0.1705
Panel B: Physical investment, $I^K/(K+N)$					
q	0.0018*** (0.0002)	0.0149*** (0.0019)	0.0333*** (0.0021)	0.0340*** (0.0021)	0.0381*** (0.0022)
Observations	105,979	105,979	105,979	105,979	105,979
R^2	0.5072	0.5410	0.5503	0.5464	0.5491
Within R^2	0.0228	0.0898	0.1082	0.1005	0.1059
Panel C: Intangible investment, $I^N/(K+N)$					
q	0.0029*** (0.0002)	0.0154*** (0.0008)	0.0273*** (0.0015)	0.0285*** (0.0016)	0.0299*** (0.0017)
Observations	105,979	105,979	105,979	105,979	105,979
R^2	0.7795	0.7971	0.7861	0.7850	0.7824
Within R^2	0.1138	0.1848	0.1405	0.1363	0.1258

Table IA.7: Investment and q : value components, scaled by total capital

This table repeats Table 9 with every investment rate and value component scaled by total capital $K+N$ rather than by the capital stock of the margin being explained, and adds the total-investment margin. Panel A reports the investment regressions. Columns (1)–(2) explain total investment $I/(K+N)$, columns (3)–(5) physical investment $I^K/(K+N)$, and columns (6)–(8) intangible investment $I^N/(K+N)$; all rates are measured one year ahead. Columns (3) and (6) use the components generated by the corresponding capital stock, columns (1), (4), and (7) use all four components, and columns (2), (5), and (8) add total q . Panel B decomposes the within- R^2 of columns (1), (4), and (7) into Shapley shares, which add to 100. All specifications include firm and year fixed effects; standard errors in Panel A are two-way clustered by firm and year.

	Total, $I/(K+N)$		Physical, $I^K/(K+N)$			Intangible, $I^N/(K+N)$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Investment regressions								
V^{phy}	0.0739*** (0.0030)	0.0429*** (0.0033)	0.0513*** (0.0024)	0.0509*** (0.0024)	0.0408*** (0.0025)		0.0212*** (0.0016)	0.0008 (0.0015)
$V^{\text{prod-int}}$	0.0413*** (0.0060)	-0.0007 (0.0049)		0.0096*** (0.0028)	-0.0041 (0.0026)	0.0278*** (0.0040)	0.0303*** (0.0041)	0.0027 (0.0032)
$V^{\text{mkp-int}}$	0.0481*** (0.0138)	0.0582*** (0.0129)		-0.0218** (0.0082)	-0.0185** (0.0079)	0.1225*** (0.0092)	0.0730*** (0.0087)	0.0797*** (0.0082)
$V^{\text{mkp-non}}$	0.0514*** (0.0055)	0.0329*** (0.0053)		0.0365*** (0.0036)	0.0305*** (0.0036)		0.0132*** (0.0030)	0.0010 (0.0029)
Total q		0.0190*** (0.0016)			0.0062*** (0.0008)			0.0125*** (0.0008)
Observations	105,979	105,979	105,979	105,979	105,979	105,979	105,979	105,979
R^2	0.6180	0.6325	0.5579	0.5627	0.5661	0.7791	0.7884	0.8016
Within R^2	0.1866	0.2173	0.1233	0.1328	0.1396	0.1125	0.1498	0.2029
Panel B: Shapley decomposition of within- R^2 (%)								
	Total, $I/(K+N)$		Physical, $I^K/(K+N)$			Intangible, $I^N/(K+N)$		
V^{phy}	68.09		85.23			35.00		
$V^{\text{prod-int}}$	11.72		2.96			29.26		
$V^{\text{mkp-int}}$	14.76		5.27			32.50		
$V^{\text{mkp-non}}$	5.43		6.54			3.24		
Total within R^2	0.1866		0.1328			0.1498		
Observations	105,979		105,979			105,979		

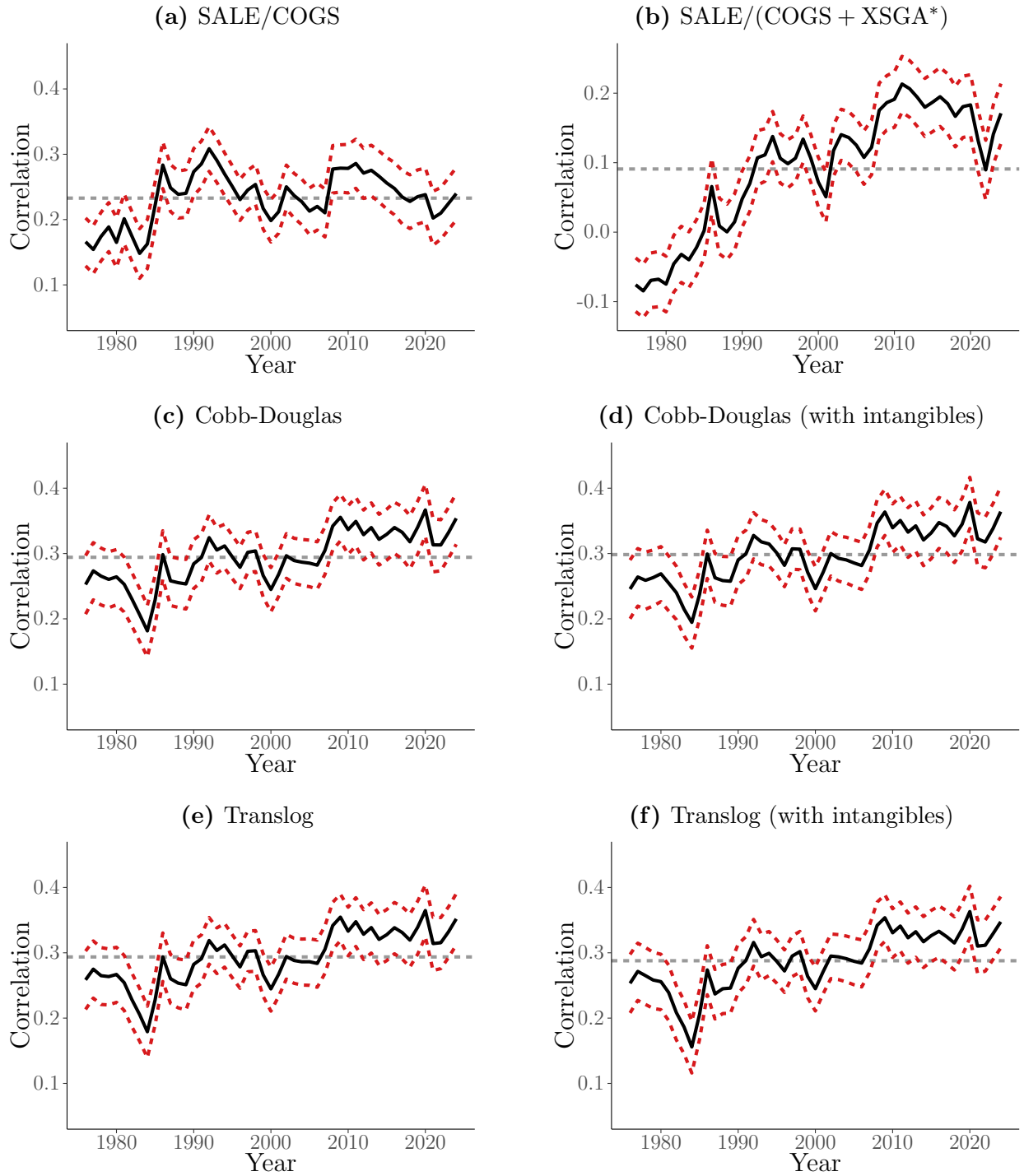


Figure IA.1: Correlation between intangibles and markup: alternative specifications (1976–2024)

This figure repeats Panel (a) of Figure 1 for all six markup measures. The specification is identical; see the note to Figure 1.

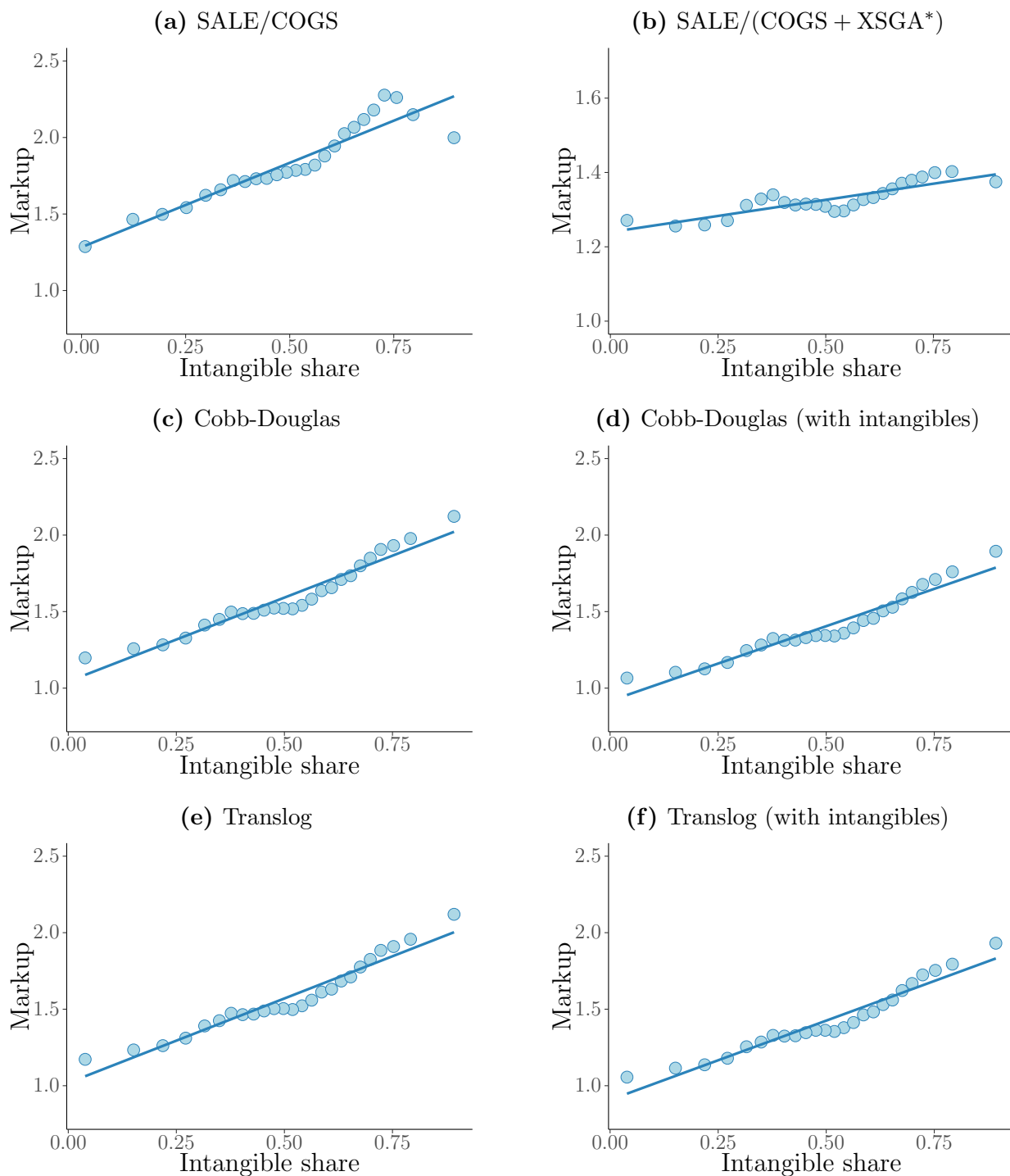


Figure IA.2: Binned scatter plots of markup against intangible share: alternative specifications (1976–2024)

This figure repeats Panel (b) of Figure 1 for all six markup measures. The specification is identical; see the note to Figure 1.

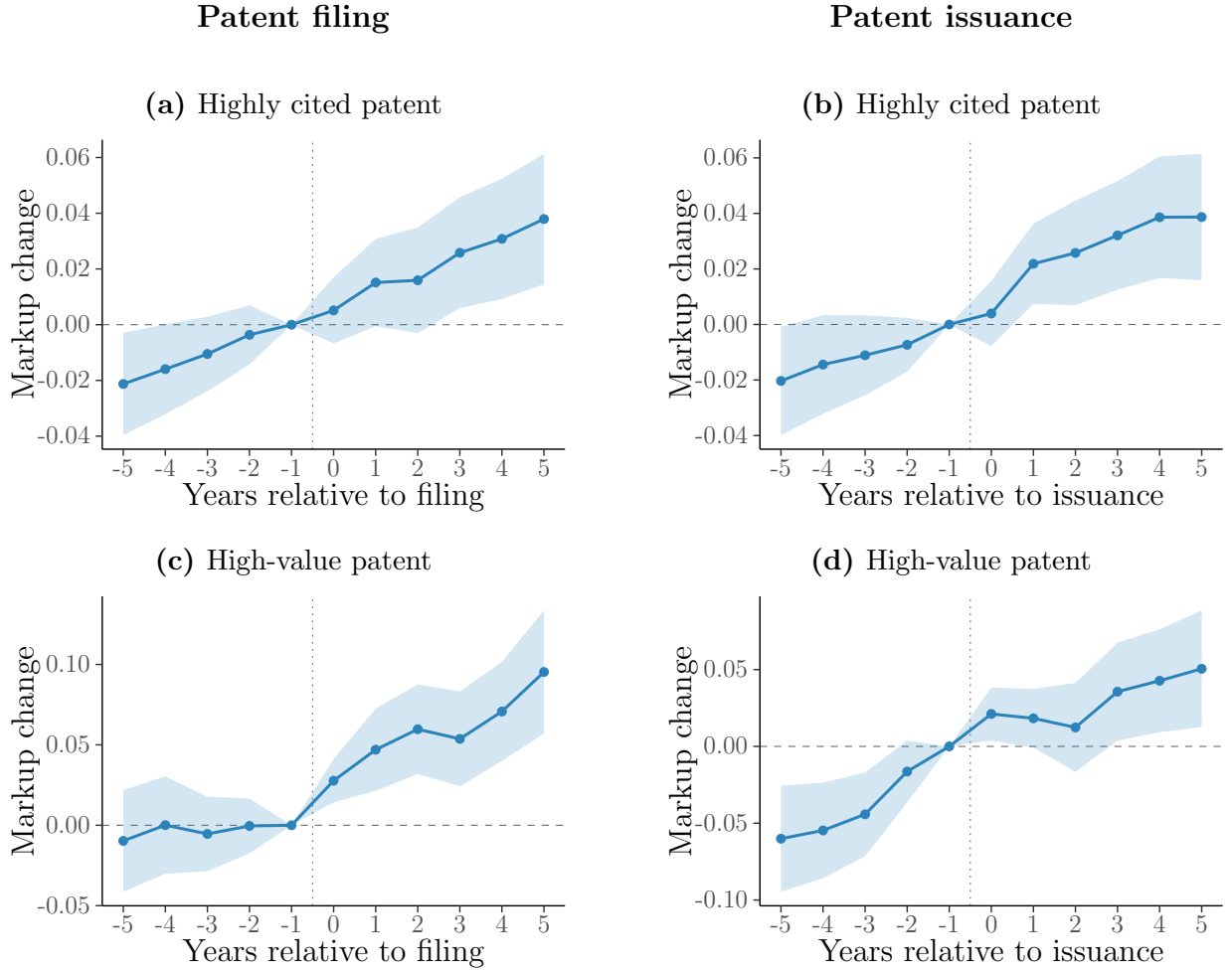


Figure IA.3: Event studies under Cobb-Douglas with intangibles: stacked difference-in-differences

This figure reproduces Figure 2 using a stacked difference-in-differences design in place of the first-event specification for important patent filings (left column) and issuances (right column), using the “Cobb-Douglas with intangibles” markup measure. Panels (a) and (b) define important patents by forward citations in the top 10% within their Fama-French 48 industry and year; Panels (c) and (d) instead use the Kogan et al. (2017) stock-market value of the patent. Each firm-year with an important patent forms a separate event stack: the treated firm is followed over $\tau \in [-5, +5]$, and controls are firms in the same Fama-French 48 industry with no important patent within that window. Stacks are pooled with event-specific firm and year fixed effects, coefficients are normalized relative to the year before the event ($\tau = -1$), and treated firms are required to have markup data in every pre-event year $\tau \in \{-5, \dots, -1\}$. Shaded areas are 95% confidence intervals, with standard errors two-way clustered by event and firm.

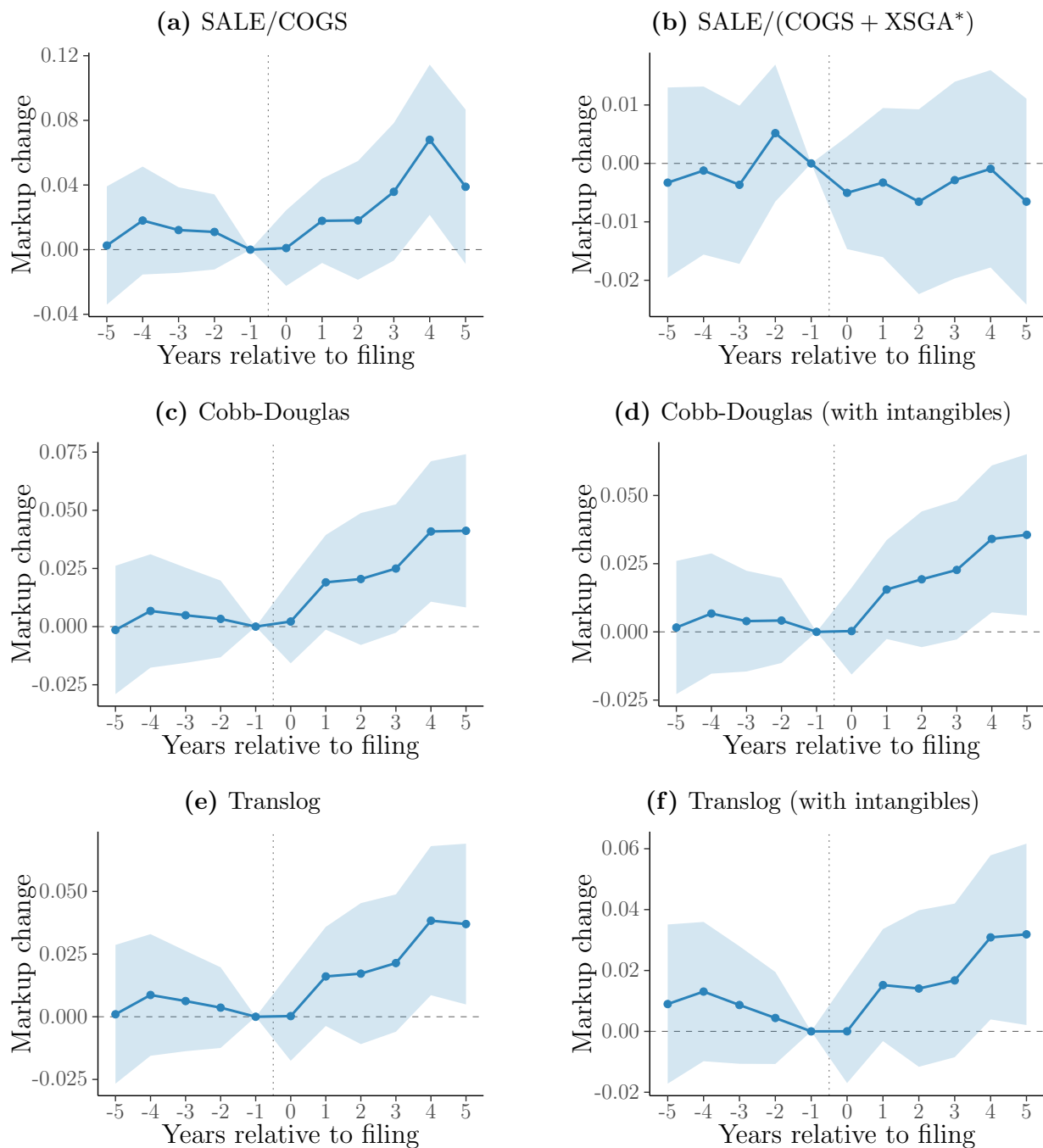


Figure IA.4: Event study around patent filing: highly cited patents

This figure repeats the event study of Figure 2 (Panels (a) and (b)) around patent filing, for all six markup measures. The specification is identical; see the note to Figure 2.

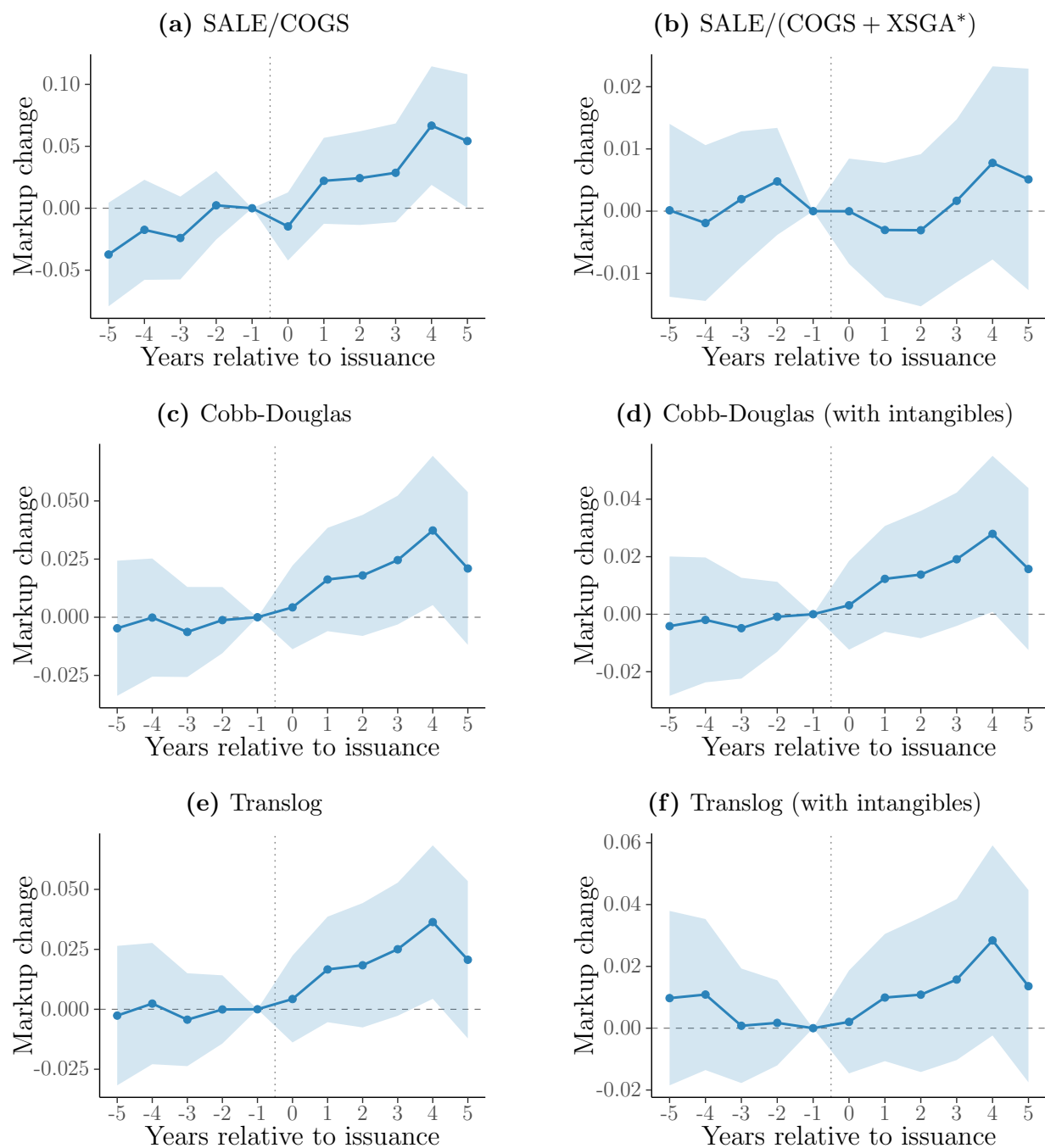


Figure IA.5: Event study around patent issuance: highly cited patents

This figure repeats the event study of Figure 2 (Panels (a) and (b)) around patent issuance, for all six markup measures. The specification is identical; see the note to Figure 2.

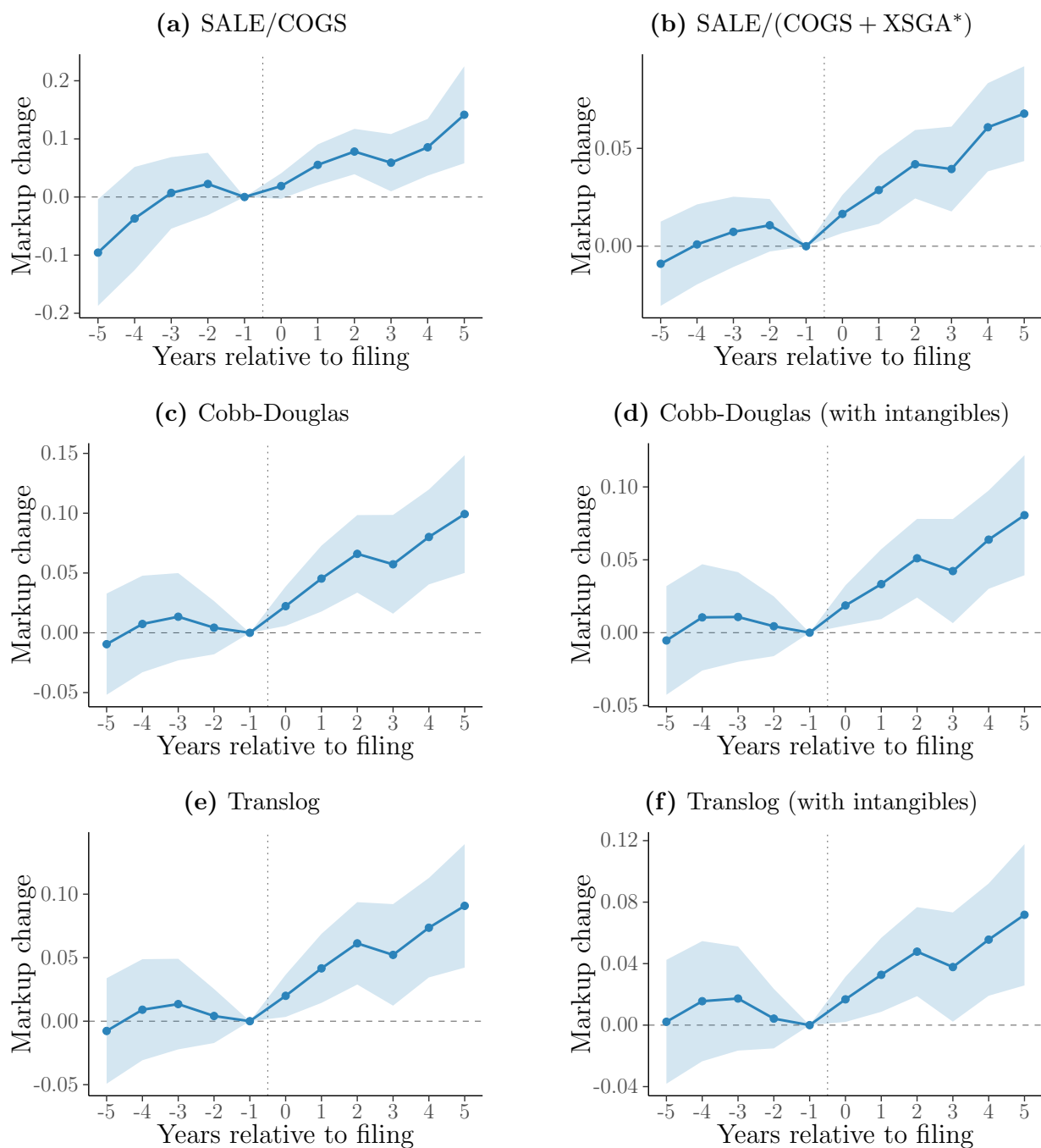


Figure IA.6: Event study around patent filing: high-value patents

This figure repeats the event study of Figure 2 (Panels (c) and (d)) around patent filing, for all six markup measures, where important patents are defined by the Kogan et al. (2017) stock-market value rather than forward citations. The specification is identical; see the note to Figure 2.

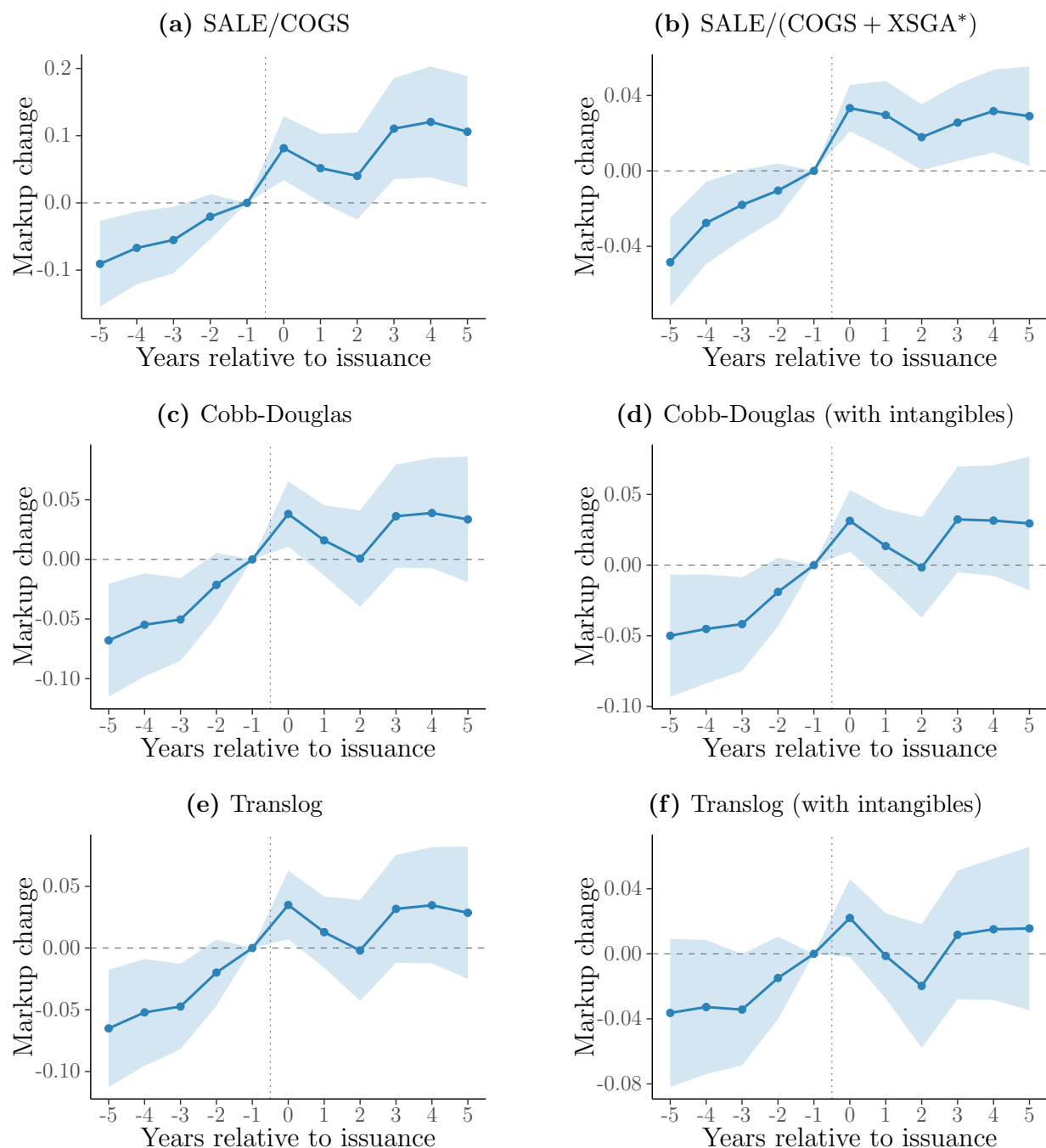


Figure IA.7: Event study around patent issuance: high-value patents

This figure repeats the event study of Figure 2 (Panels (c) and (d)) around patent issuance, for all six markup measures, where important patents are defined by the Kogan et al. (2017) stock-market value rather than forward citations. The specification is identical; see the note to Figure 2.

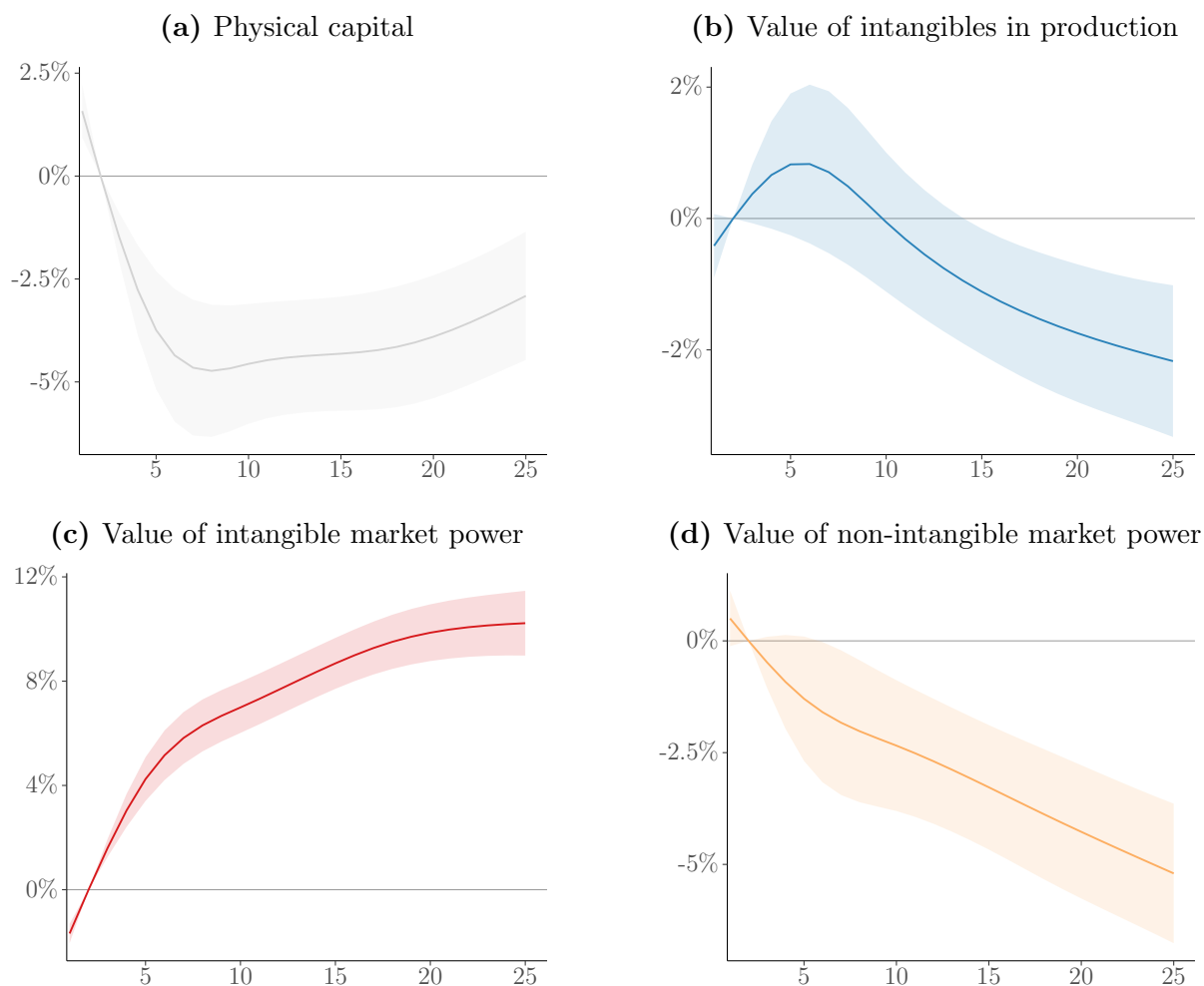


Figure IA.8: Life cycle changes

This figure plots the evolution of each value component's share over a firm's life cycle. For each component, I regress the ratio of the component to total model-implied value on firm age splines with year and cohort fixed effects, restricting the sample to firms with at least 15 years of data. The solid line shows the estimated coefficients and the shaded area represents the 95% confidence interval.